

CMPE 256 – Recommender Systems



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University.



About Me

1. CEO, Data Analytics Startup
2. CTO, Sanjeevani EHR
3. Principal, Advanced Analytics, Microsoft
4. Principal, Deloitte and Touché
5. Sr. Program Manager, GE Healthcare
6. Product Manager, Cisco Systems
7. Architect & Project Manager, St. Jude Medical,
8. Sr. Software Engineer, Bell Labs,
9. Software Engineer, Tata Elxsi

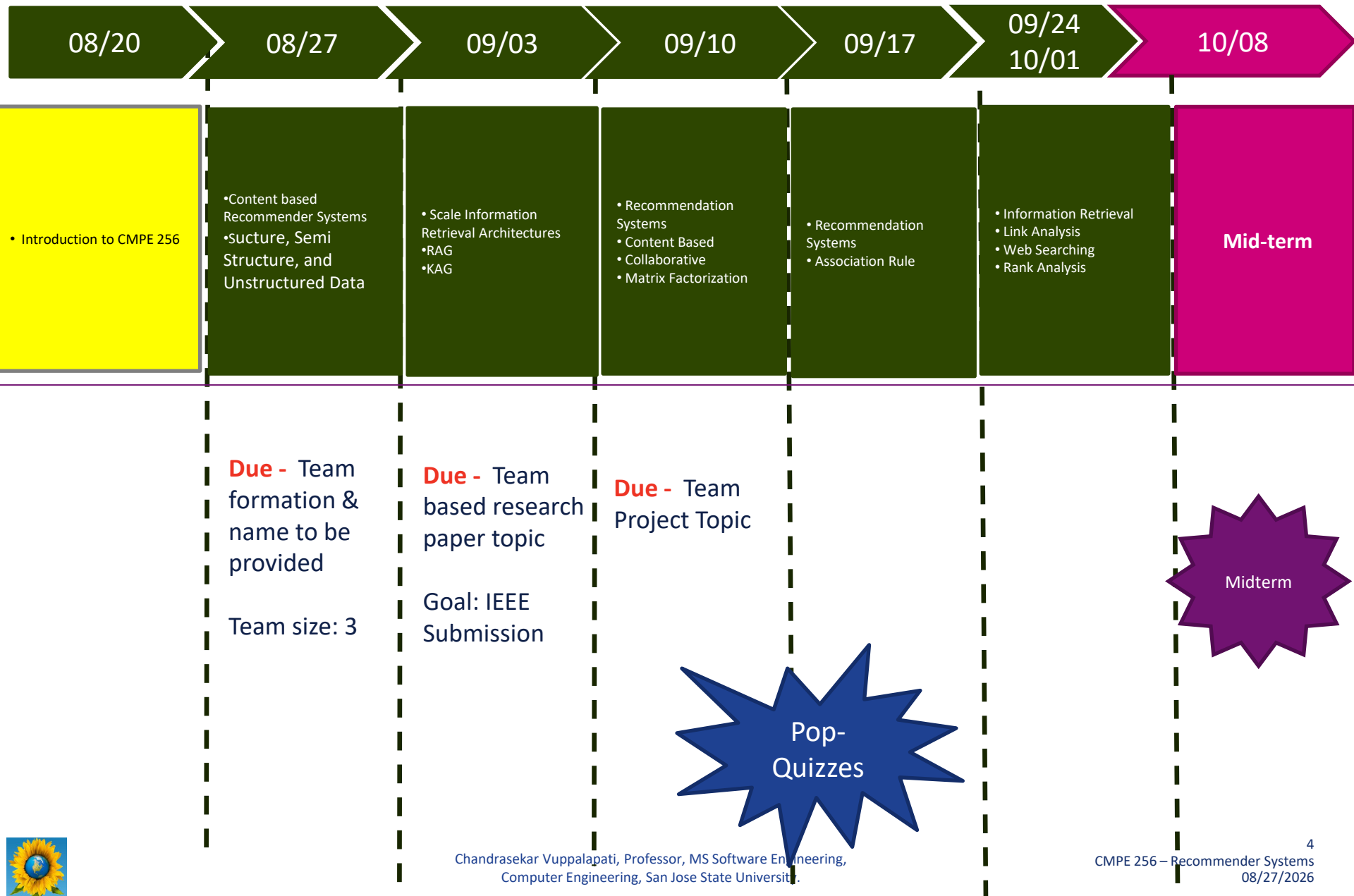


About Me

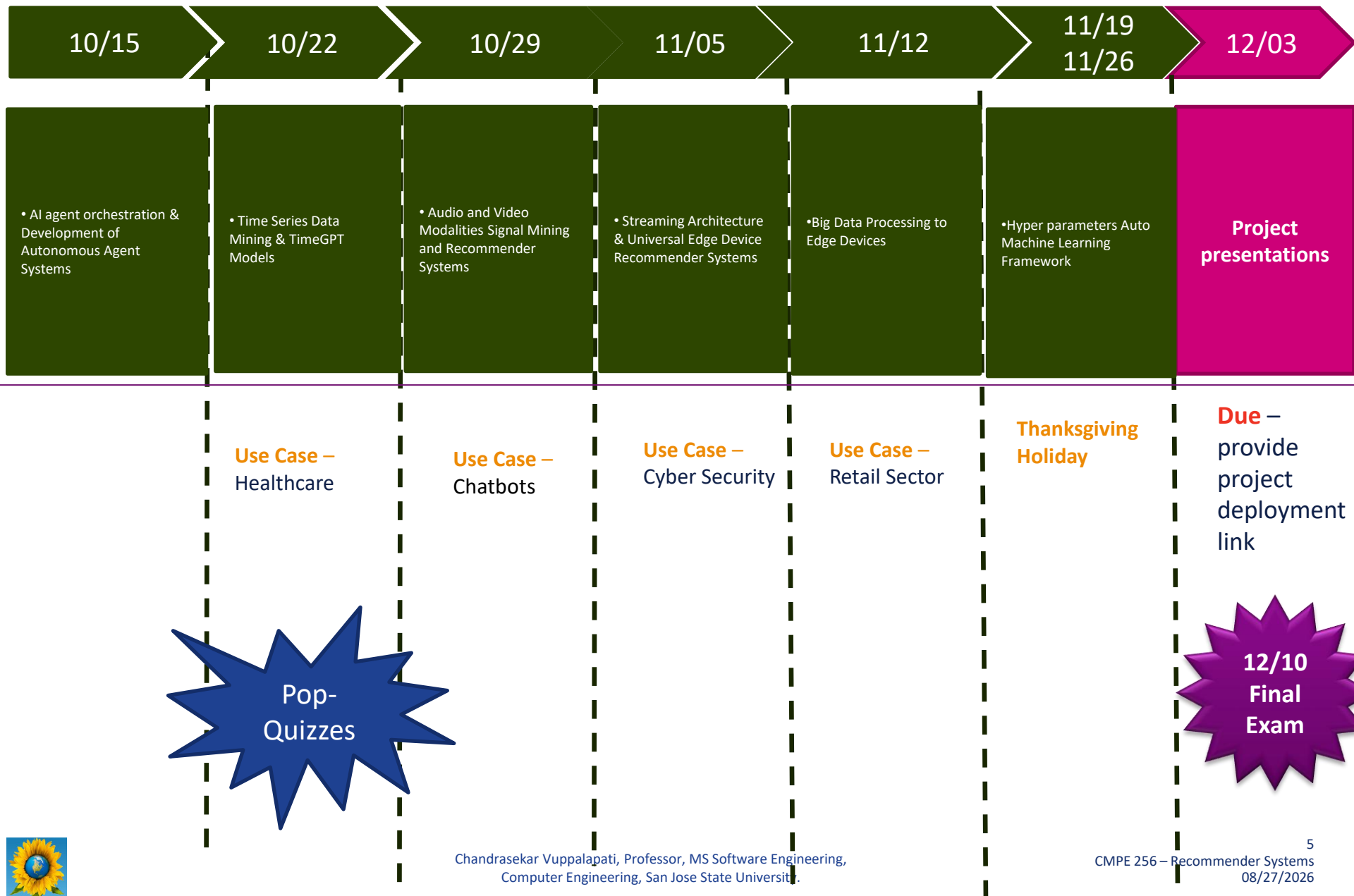
- Entrepreneur
 - Data Science Companies in Bay Area
 - IoT Sensor Manufacturer (SF, South Korea & India)
 - Healthcare Analytics Company (India)
- Principal & Director of Cyber Security & Assurance
- Principal AI/Cognitive Specialist - Big 4 Audit Company
- Visiting Professor – Cisco Cyber Security Program
- Professor, San Jose State University
- Senior Vice President (SVP)
- CTO
- Author
- **Father of Two Wonderful Daughters**



Class Roadmap



Class Roadmap



Agenda

1. Course Objectives
2. Things You will learn
3. What is Data Mining?



Things You will Learn

- Discuss and apply recommender-system concepts and techniques using state-of-the-art technologies and platforms.
- Design and implement vector-based search/retrieval recommenders (dense/sparse embeddings, ANN indices, hybrid retrieval).
- Prepare, preprocess, analyze, and mine unstructured data (web/app logs, text, audio) to build personalized services, selecting appropriate techniques and algorithms for each modality.
- Apply web mining & NLP to extract signals from social platforms, blogs, and review sites for personalization.
- Explain how web search engines index and rank content (information retrieval, SEO, graph-mining algorithms) and relate these methods to recommendation.



Things You will Learn

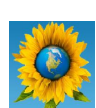
- Explain how web search engines index and rank content (information retrieval, SEO, graph-mining algorithms) and relate these methods to recommendation.
- Use prompt engineering to build advanced recommenders over transactional, structured, and unstructured datasets.
- Explain and architect agentic recommenders, including multi-agent orchestration, Agent-to-Agent (A2A) protocols, and state models for short- and long-term memory within the recommendation workflow.
- Effectively present and communicate the knowledge they have acquired in the course.

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Course Modules

- Recommendation Systems
- Information retrieval and Web Search
- Link Analysis
- Generative AI Models and Large Language Models
- Time Series Modeling & TimeGPT
- Multi Modalities
- Audio, Video Data & Recommendation
- RAG & KAG Design Patterns & ChatGPT
- Agents and Mult Agentic Architecture
- Copilot Architectures



Evaluation

- In-class assignments (5%)
- Individual home-work assignments (5%)
- Pop Quizzes (10%)
- Team Research Paper / Hackathon (15%)
- Team Project (20%)
- Exams
 - Mid-term (15%)
 - Final, comprehensive, (30%)



In-class assignments, home-work, Pop Quizzes

- White board solution
- Write high-level pseudo algorithm
- Quizzes – multiple choices, fill in the blanks
- Home assignments
 - Programming samples (on WEKA)
 - Programming Samples on R (if permitted)



Team Project

- Goal: Gain hands-on experience by designing/implementing a data/web algorithm or application
 - Group of 2-3 people
 - Your final grade will be posted based on individual contribution & assessment in ALL project deliverables
 - Project
 - Develop a project of your choice using AWS and Apache Mahout or AWS and R



Department Policy

“All non-proctored report (or similarly sized) assignments in courses where some of the final grade depends on prose writing will be submitted to Turnitin.com.”



University Policies

- **Academic Integrity Statement**

- ☐ Your own commitment to learning, as evidenced by your enrollment at San Jose State University, and the University's Integrity Policy, require you to be honest in all your academic course work. Faculty members are required to report all infractions to the office of Judicial Affairs." The policy on academic integrity can be found at:
 - http://sa.sjsu.edu/judicial_affairs/index.html
 - Incidents of violation of academic integrity are taken seriously, and will be dealt with in accordance with the SJSU policy on Academic Integrity and other related university, college, or departmental policies and regulations.



Academic Dishonesty

- 1.1 CHEATING
- San José State University defines cheating as the act of obtaining or attempting to obtain credit for academic work through the use of any dishonest, deceptive, or fraudulent means. Cheating includes:
 - 1.1.1 Copying, in part or in whole, from another's test or other evaluation instrument including homework assignments, worksheets, lab reports, essays, summaries, quizzes, etc.;
 - 1.1.2 Submitting work previously graded in another course without prior approval by the course instructor or by departmental policy;
 - 1.1.3 Submitting work simultaneously presented in two courses without prior approval by both course instructors or by the department policies of both departments;
 - 1.1.4 Using or consulting sources, tools or materials prohibited by the instructor prior to, or during an examination;
 - 1.1.5 Altering or interfering with the grading process;
 - 1.1.6 Sitting for an examination by a surrogate, or as a surrogate;
 - 1.1.7 Any other act committed by a student in the course of their academic work that defrauds or misrepresents, including aiding others in any of the actions defined above.



Academic Dishonesty

- 1.2 PLAGIARISM
- San José State University defines plagiarism as the act of representing the work of another as one's own without giving appropriate credit, regardless of how that work was obtained, and submitting it to fulfill academic requirements.
- Plagiarism includes:
 - 1.2.1 Knowingly or unknowingly incorporating the ideas, words, sentences, paragraphs, or parts of, or the specific substance of another's work, without giving appropriate credit, and representing the product as one's own work;
 - 1.2.2 Representing another's artistic/scholarly works such as musical compositions, computer programs, photographs, paintings, drawing, sculptures, or similar works as one's own.



Pledge

- You will sign the pledge along with the prerequisite agreement next week. Take some time and read the honesty pledge carefully.
 - Available from class web site
- “I have read and understand the university definition of cheating and plagiarism.
- I will not copy (“cut and paste”) *any* material from *any* source without proper attribution.
- I will not discuss *any* assignment that is part of the course grade with *anybody* without prior approval from the instructor.
- I understand that there is no make-up opportunity for assignments where breaches of academic integrity have occurred.
- I understand that breaches of academic integrity will result in a failing grade in the course, as well as possible referral for additional administrative sanctions.
- I will ask the instructor for guidance if I have any questions about how the above is to be understood.”



University Policies

- You are responsible for understanding the policies and procedures about add/drops, academic renewal, withdrawals, incompletes, classroom behavior, and other policies described in the catalog. Please read your catalog thoroughly.
- **Campus policy in compliance with the Americans with Disabilities Act**

If you need course adaptations or accommodations because of a disability, or if you need to make special arrangements in case the building must be evacuated, please make an appointment with me as soon as possible, or see me during office hours. Presidential Directive 97-03 at

http://www.sjsu.edu/president/docs/directives/PD_1997-03.pdf requires that students with disabilities requesting accommodations must register with the [Accessible Education Center](#) (AEC) at <http://www.sjsu.edu/aec> to establish a record of their disability.



University Policies

Consent for Recording of Class and Public Sharing of Instructor Material

- According to [University Policy SI2-7](http://www.sjsu.edu/senate/docs/SI2-7.pdf), <http://www.sjsu.edu/senate/docs/SI2-7.pdf>, students need to obtain instructor's permission to record the course. Such permission allows the recordings to be used for your private, study purposes only. The recordings are the intellectual property of the instructor; you have not been given any rights to reproduce or distribute the material. This should happen on a lecture-by-lecture basis, in writing, at least one day prior to the lecture.
- Course material developed by the instructor is the intellectual property of the instructor and cannot be shared publicly without his/her approval. You may not publicly share or upload instructor generated material for this course such as exam questions, lecture notes, or homework solutions without instructor consent.





Data Mining

Google Wallet Creators Raise \$7M From Eric Schmidt, Khosla To Bring Personalization And Analytics To In-Store Commerce

Posted Nov 19, 2013 by [Leena Rao \(@leenarao\)](#)

1 Like 176 Tweet 372 Share 114 ▼

Source: <http://techcrunch.com/2013/11/19/google-wallet-creators-raise-7m-from-eric-schmidt-khosla-to-bring-personalization-and-analytics-to-in-store-commerce/>



While Amazon has been at the forefront of providing a personalized experience in online commerce, in-store retailers are still behind in using data in compelling ways to make the purchasing and shopping experience better in physical retail. **Index**, the new personalization and analytics company from the original members of the Google Wallet team, is hoping to change

this. Today, the startup is announcing new funding and more details on the company's product. Index has raised \$7 million in a Series A round led by Eric Schmidt's Innovation Endeavors, Khosla Ventures, AIMCo and 819 Capital.

At its core, Index wants to brick-and-mortar retailers leverage their physical stores as a competitive differentiator. **Founded by former Google Wallet executives** Marc Freed-Finnegan and Jonathan Wall, Index offers top-tier, brick and mortar retailers (as the Macy's or the Best Buys of the world) a software integration that plugs into their point-of-sale

Index

FOUNDED
March 2012

TOTAL FUNDING
Not available

OVERVIEW

Index is building a next generation Commerce Network from the ground up. We believe that better relationships drive better business. We are transforming offline retail by strengthening the connection between retailers and their customers. Our team of engineers, designers, and business gurus work side by side with a shared vision of revolutionizing commerce.

WEBSITE

<http://www.goindex.com>

The software includes the ability to provide a linked identity and enable retailers to develop unified customer profiles and deliver truly personalized service. **Data mining is another big part of the technology—with the hope of going beyond just targeting by checkout basket.**

Decide.com Raises \$8M Series C For Its Price Predictions Engine; Will Expand To More Categories, Improve Data Mining & Mobile Apps

Source: <http://techcrunch.com/2013/03/22/decide-com-raises-8m-series-c-for-its-price-predictions-engine-will-expand-to-more-categories-improve-data-mining-mobile-apps/>

Posted Mar 22, 2013 by [Sarah Perez \(@sarahintampa\)](#)

0 Like 0 Tweet 204 Share 0



Shopping and comparison engine [Decide.com](#), which offers suggestions as to not just what to buy but also *when*, is announcing today that it has raised \$8 million in Series C funding, in a round led by Vulcan Capital. Also

participating were existing investors, Maveron and Madrona Venture Group.

As a part of the new funding, Steve Hall from Vulcan Capital is joining Decide.com's board. The company also recently added another board member as well: Dawn Lepore, former CEO of Drugstore.com.

Decide.com

FOUNDED
June 2011

TOTAL FUNDING
\$16.5M

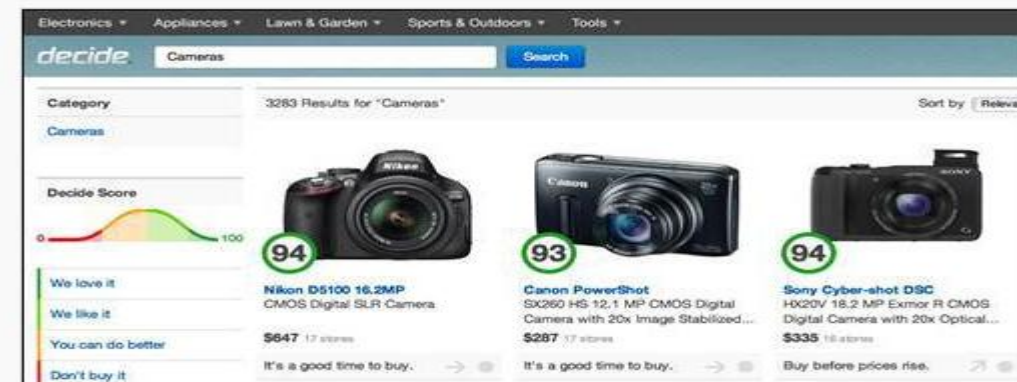
OVERVIEW

Backed by early investors in Amazon, Google and eBay, Decide is a team of Seattle data scientists committed to helping consumers easily find the highest rated products and the best times to buy based on billions of data points from across the Web. Since launch in June 2011, Decide's price predictions have been 80 percent accurate and have helped uncover \$200 million in savings. The company was co-founded ...

WEBSITE

<http://www.decide.com>

[Full profile for Decide.com](#)



Decide.com [initially began by targeting consumer electronics](#), but later expanded into other product categories. Today the company covers 150 product categories across Electronics, Appliances, and Home & Garden. CEO Mike Fridgen says that the plan is to increase product coverage by 10 times in the weeks ahead, including moves into Baby & Kids, Jewelry, and Toys.

Intuit Acquires Level Up Analytics, A Consulting Firm With A Specialty In Data Analytics

Posted Oct 23, 2013 by [Alex Williams \(@alexwilliams\)](#)

1  Like  Tweet 420  Share 190

Source: <http://techcrunch.com/2013/10/23/intuit-acquires-level-analytics-a-consulting-firm-with-a-specialty-in-data-analytics/>



Intuit acquired 14 employees from Level Up Analytics: three founders and 11 scientists, architects, and engineers. In a blog post, Level Up Analytics wrote that its clients include companies in the financial services, advertising, mobile, payments and social networking fields.

The Level Up Analytics team will help build products on top of Intuit's collective data of more than 45 million customers that ranges from individual purchases and spending habits to business inventories, transactions, and trends.

The company's web site explains that its core business is to "weave techniques from machine learning, data mining, predictive modeling, graph and social network analysis, natural language processing and software engineering into its products and analyses." Its clients have used its services to churn analysis, information extraction, data standardization, recommendation systems, data modeling and a host of other services.

Intuit has made several acquisitions this year. In August, the company acquired tax software startup GoodApril.

Intuit has acquired [Level Up Analytics](#), a Mountain View-based consulting company that focuses on data science, big data and analytics. Terms of the deal were not disclosed. All the employees from the Level Up Analytics team will join Intuit.

In a statement, Intuit described the deal as an "acqui-hire," its fifth this year. The company did not provide the names of the other consulting firms.

Big Data Characteristics

The sheer volume of data being stored today is exploding.

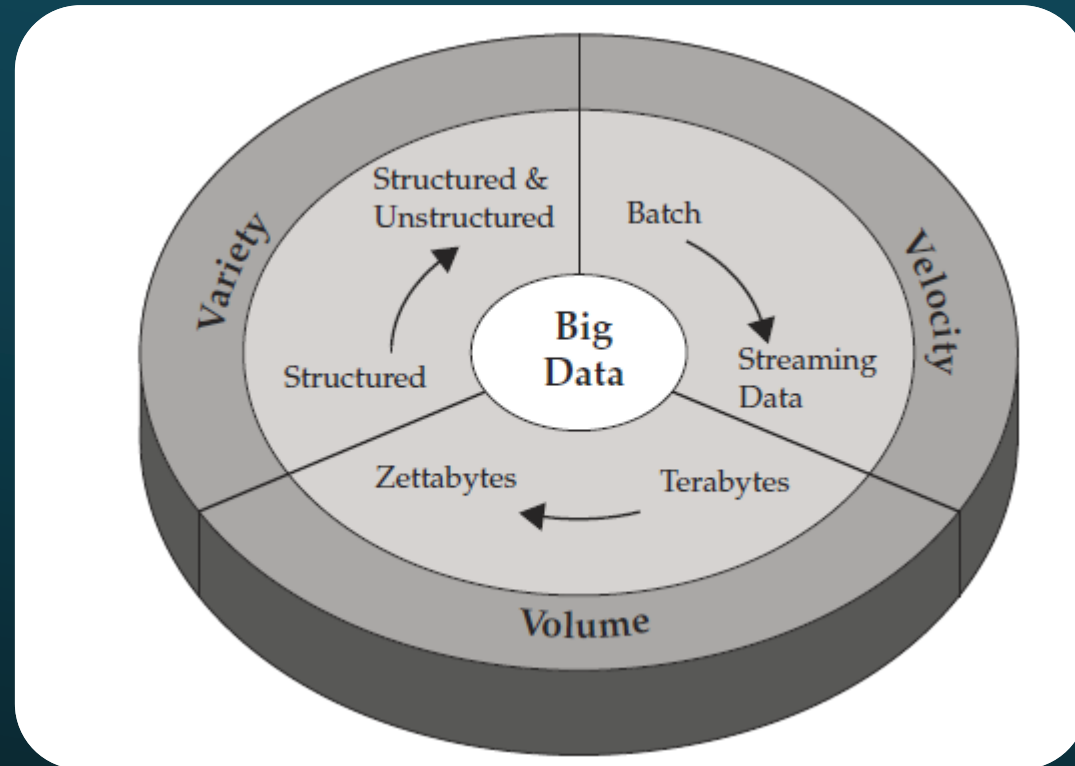
Year	Data
2000	800,000 petabytes (PB) of data were stored in the world.
2020	35 zettabytes (ZB)



More than 7 terrabytes (TB) of data every day



> 10 terrabytes (TB) of data



In Order to Realize New Opportunities, You Need to Think Beyond Traditional Sources of Data

Transactional and Application Data



- Volume
- Structured
- Throughput

Machine Data



- Velocity
- Semi-structured
- Ingestion

Social Data



- Variety
- Highly unstructured
- Veracity

Enterprise Content



- Variety
- Highly unstructured
- Volume

Why Data Driven Business?



Data driven decisions are better decisions – its as simple as that. Using big data enables mangers to decide on the basis of evidence rather than intuition. For that reason it has the potential to revolutionize management

**Harvard Business Review
October 2012**

Every Industry can Leverage Big Data and Analytics



Banking

- Optimizing Offers and Cross-sell
- Customer Service and Call Center Efficiency
- Fraud Detection & Investigation
- Credit & Counterparty Risk



Insurance

- 360° View of Domain or Subject
- Catastrophe Modeling
- Fraud & Abuse
- Producer Performance Analytics
- Analytics Sandbox



Telco

- Pro-active Call Center
- Network Analytics
- Location Based Services



Energy & Utilities

- Smart Meter Analytics
- Distribution Load Forecasting/Scheduling
- Condition Based Maintenance
- Create & Target Customer Offerings



Media & Entertainment

- Business process transformation
- Audience & Marketing Optimization
- Multi-Channel Enablement
- Digital commerce optimization



Retail

- Actionable Customer Insight
- Merchandise Optimization
- Dynamic Pricing



Travel & Transport

- Customer Analytics & Loyalty Marketing
- Predictive Maintenance Analytics
- Capacity & Pricing Optimization



Consumer Products

- Shelf Availability
- Promotional Spend Optimization
- Merchandising Compliance
- Promotion Exceptions & Alerts



Government

- Civilian Services
- Defense & Intelligence
- Tax & Treasury Services



Healthcare

- Measure & Act on Population Health Outcomes
- Engage Consumers in their Healthcare



Automotive

- Advanced Condition Monitoring
- Data Warehouse Optimization
- Actionable Customer Intelligence



Chemical & Petroleum

- Operational Surveillance, Analysis & Optimization
- Data Warehouse Consolidation, Integration & Augmentation
- Big Data Exploration for Interdisciplinary Collaboration



Aerospace & Defense

- Uniform Information Access Platform
- Data Warehouse Optimization
- Airliner Certification Platform
- Advanced Condition Monitoring (ACM)



Electronics

- Customer/ Channel Analytics
- Advanced Condition Monitoring



Life Sciences

- Increase visibility into drug safety and effectiveness

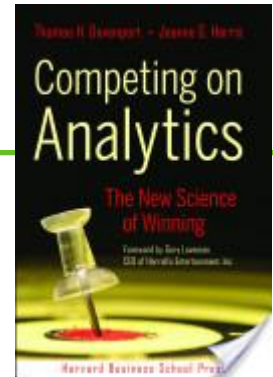
Most Requested Uses of Big Data

- Log Analytics & Storage
- Fusion of multi-INT (multi-formats)
- RFID Tracking & Analytics
- Fraud Detection & Modeling
- Risk Modeling & Management
- 360° View of a Person, Place, or Thing
- Warehouse Extension (case patterns)
- Email / Call Center Transcript Analysis
- Call Detail Record Analysis
- IBM Watson



[https://www-950.ibm.com/events/wwe/grp/grp004.nsf/vLookupPDFs/Steve%20Stennet's%20Presentation/\\$file/Steve%20Stennet's%20Presentation.pdf](https://www-950.ibm.com/events/wwe/grp/grp004.nsf/vLookupPDFs/Steve%20Stennet's%20Presentation/$file/Steve%20Stennet's%20Presentation.pdf)

Competing on Analytics



THINGS YOU CAN COUNT ON

Analytics competitors make expert use of statistics and modeling to improve a wide variety of functions. Here are some common applications:

FUNCTION	DESCRIPTION	EXEMPLARS
Supply chain	Simulate and optimize supply chain flows; reduce inventory and stock-outs.	Dell, Wal-Mart, Amazon
Customer selection, loyalty, and service	Identify customers with the greatest profit potential; increase likelihood that they will want the product or service offering; retain their loyalty.	Harrah's, Capital One, Barclays
Pricing	Identify the price that will maximize yield, or profit.	Progressive, Marriott
Human capital	Select the best employees for particular tasks or jobs, at particular compensation levels.	New England Patriots, Oakland A's, Boston Red Sox
Product and service quality	Detect quality problems early and minimize them.	Honda, Intel
Financial performance	Better understand the drivers of financial performance and the effects of nonfinancial factors.	MCI, Verizon
Research and development	Improve quality, efficacy, and, where applicable, safety of products and services.	Novartis, Amazon, Yahoo

Big Data Will Impact Every Aspect of Your Business



Know Everything about your Customer

Analyze all sources of data to know your customers as individuals, from channel interactions to social media.



Run Zero-latency Operations

Analyze all available operational data and react in real-time to optimize processes. Reduce the cost of IT with new cost-effective technologies.



Innovate New Products at Speed and Scale

Capture all sources of feedback and analyze vast amounts of market and research data to drive innovation.



Instant Awareness of Fraud and Risk

Develop better fraud/risk models by analyzing all available data, and detect fraud in real-time with streaming transaction analysis.



Exploit Instrumented Assets

Monitor assets from real-time data feeds to predict and prevent maintenance issues and develop new products and services.

Discover



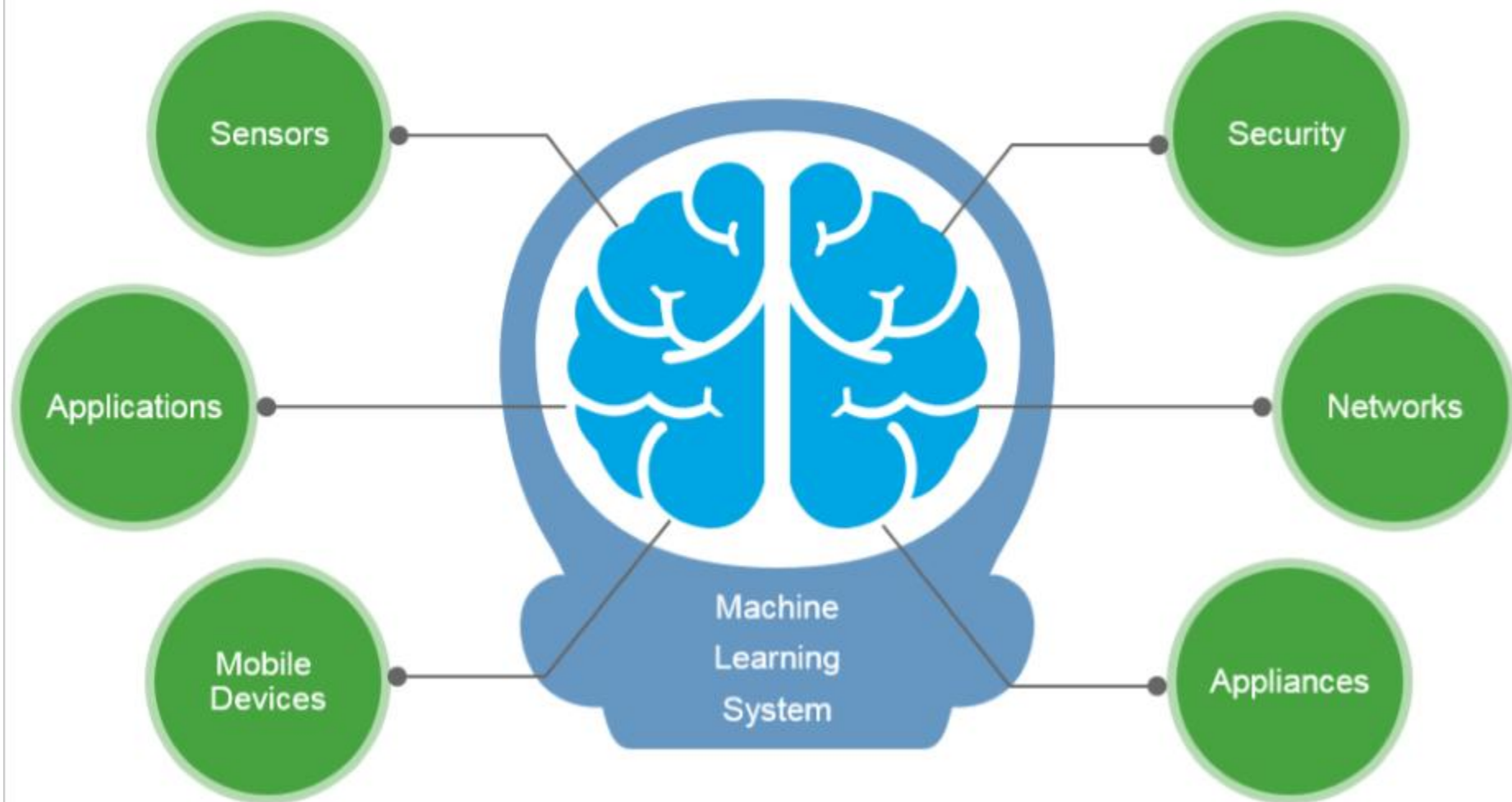
Preparing and Architecting for ML

- Machine learning (ML) — a subset of artificial intelligence (AI) — is more than a technique for analyzing data. It's a system that is fueled by data, with the ability to learn and improve by using algorithms that provide new insights without being explicitly programmed to do so.
- Preparing data for ML pipelines is challenging when end-to-end data and analytic architectures are not refined to interoperate with underlying analytic platforms..
- ML is best-suited for dealing with big data. Organizations overwhelmed with data are using multiple ML frameworks to increase operational efficiencies and achieve greater business agility.
- Technical professionals are using machine learning to add elements of intelligence to software development and IT operations (DevOps) to gain operational efficiencies.
- The ML compute and storage cluster — which is the heart of the ML system — will vary based on learning method, learning application and need for automation.

Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf



Learning from Data without being explicitly Programmed



Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf

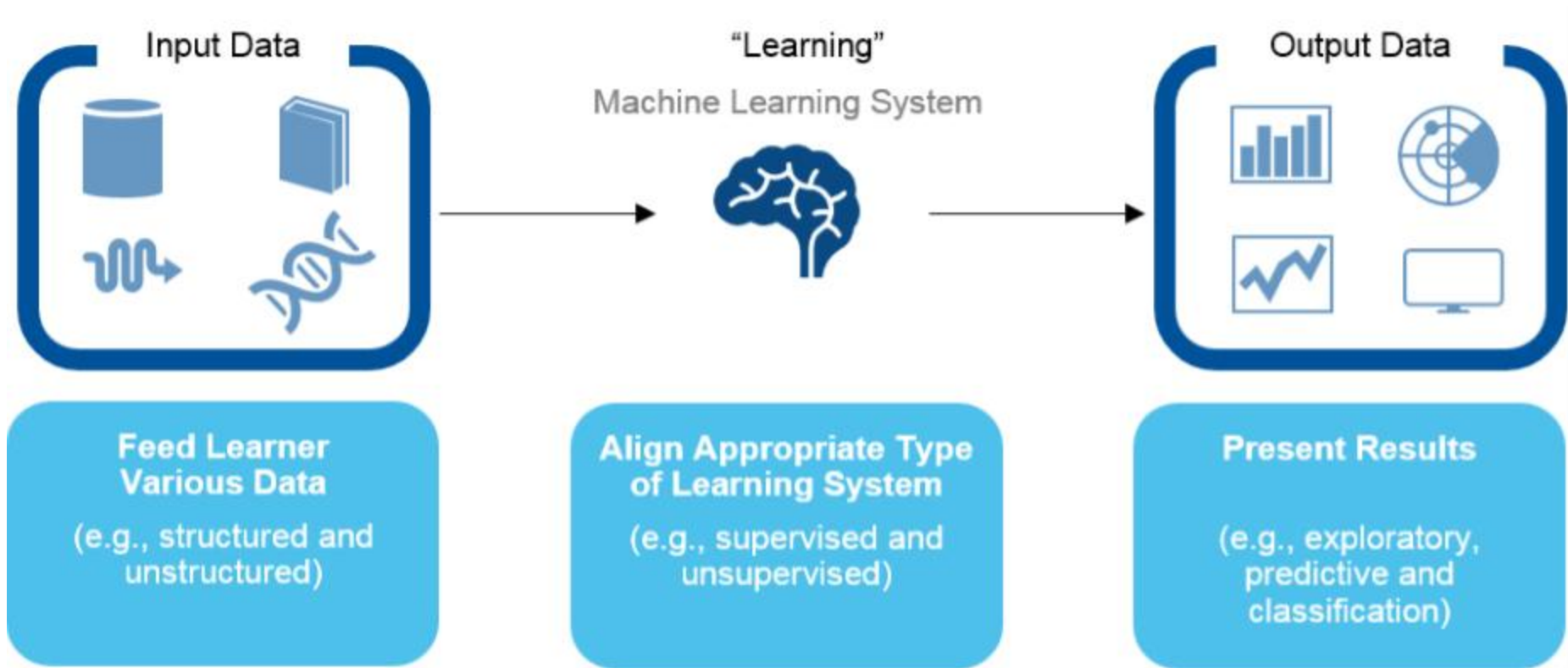
Preparing and Architecting for ML

- **Supervised learning**, where observations contain input/output pairs (aka labeled data): These sample pairs are used to "train" the machine learning system to recognize certain rules for correlating inputs to outputs. Examples include types of ML that are trained to recognize a shape based on a series of shapes in pictures.
- **Unsupervised learning**, where those labels are omitted: In this form of ML, rather than being "trained" with sample data, *the machine learning system finds structures and patterns in the data on its own*. Examples include types of ML that recognize patterns in attributes from input data that can be used to make a prediction or classify an object.
- **Reinforcement learning**, *where evaluations are given about how good or bad a certain situation* is: Examples include types of ML that enable computers to learn to play games or drive vehicles.

Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf



Basics of ML



Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf



Basics of ML

- **Learning:** Typically, the ML used for business purposes is either supervised or unsupervised in nature. Within these categories, however, there many different types of algorithms and ML routines, which can be used to accomplish different goals. Additionally, there are often different learning methods, such as "eager" and "lazy" learning methods. These methods govern how to process training data, and that governance will determine compute and storage requirements:
 - *Eager* learning methods evaluate training data and "eagerly" begin computing (for example, classification) before receiving new (test) data. They generally depend more on upfront evaluation of training data in order to compute (that is, predict) without the need for new data. As a result, eager learning methods tend to spend more time on processing the training data.
 - Lazy learning methods delay processing and data evaluation until new test data is provided — hence the term "lazy" or "lazy evaluation." As a result, lazy learning methods are often case-based, spending less time on the training data and more time on predicting.

Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf



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Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf



Learning from Data without being explicitly Programmed

Source https://www.gartner.com/binaries/content/assets/events/keywords/catalyst/catus8/preparing_and_architecting_for_machine_learning.pdf

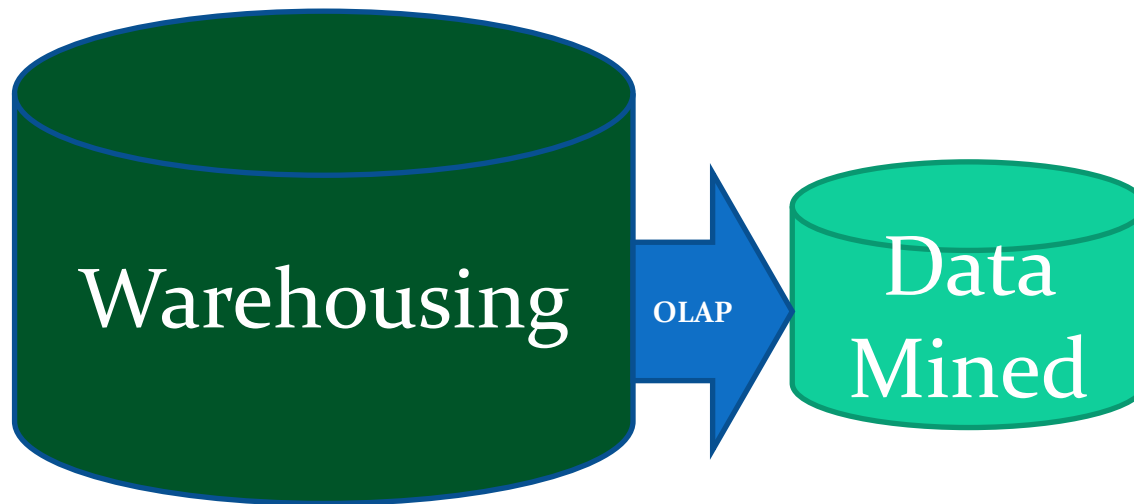


Discover



“Necessity is the Mother of Invention”

- Data explosion problem
 - Automated data collection tools and mature database technology lead to tremendous amounts of data stored in databases, data warehouses and other information repositories
- We are drowning in data, but starving for knowledge!
- Solution: Data warehousing and data mining
 - Data warehousing and on-line analytical processing (OLAP)
 - Extraction of interesting knowledge (rules, regularities, patterns, constraints) from data in large databases



Source: Data Mining Concepts and Techniques Jaiwei Han and Machelaine Kambler



What is Data Mining?

Data mining has been defined as: ***The process of discovering meaningful new correlations, patterns and trends by sifting through large amounts of data stored in repositories, using pattern recognition technologies as well as statistical and mathematical techniques.***

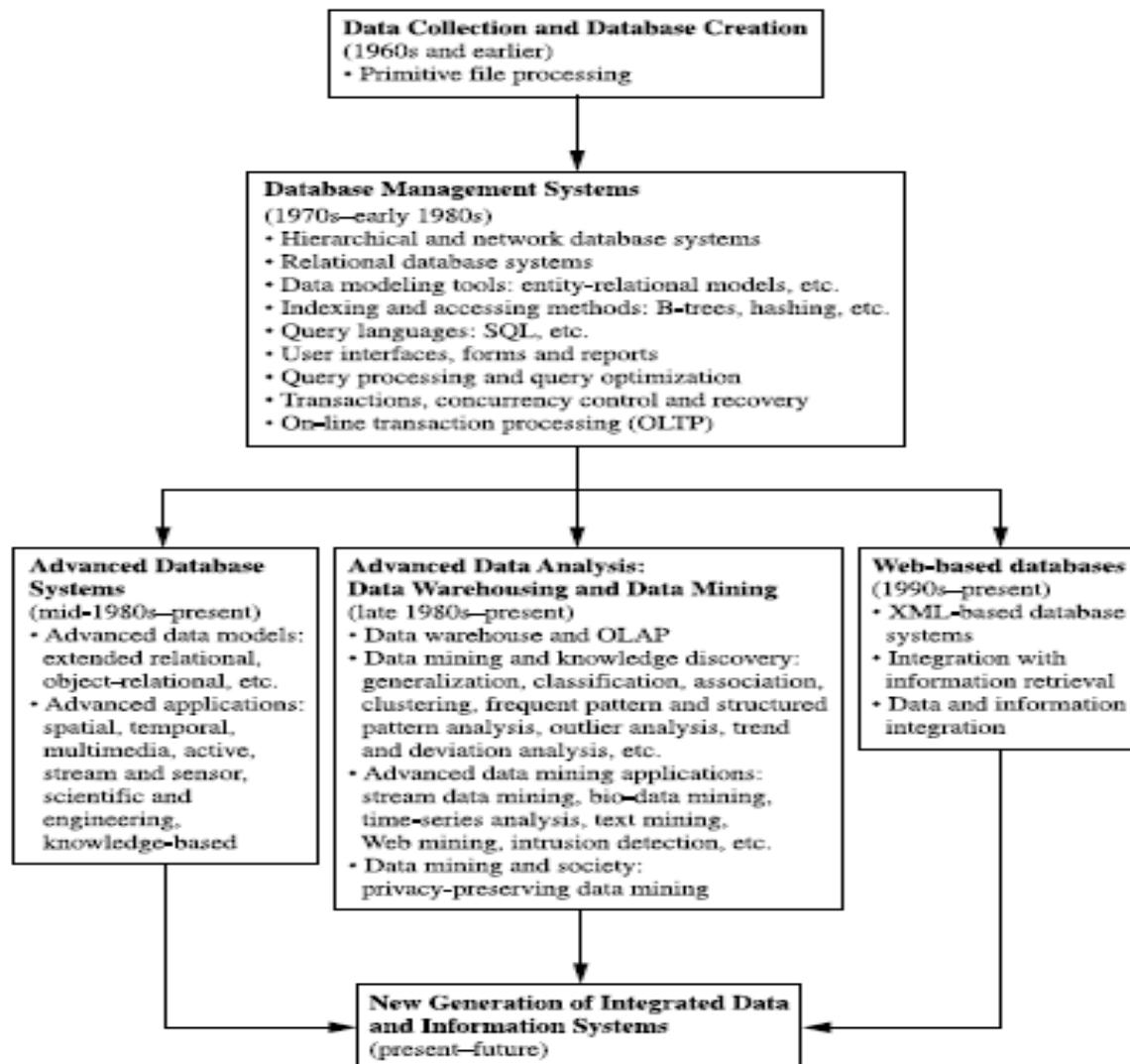
Data mining itself has also developed in the last decade. Algorithms and approaches that once demanded advanced computational resources are now available on desktop computers. Advances in database and storage technology have made mass amounts of data available in an efficient format. Finally, advances in technology surrounding the analysis of data has made it possible to include both structured and unstructured data in the data mining exercise. This has resulted in the ability to add blog, email, chat and social media data to the data mining exercise. All of this has resulted in an increased interest in, and use of, data mining.

Data mining uncovers patterns in data through a variety of predictive techniques

Source: An Introduction to Data Mining and Predictive Analytics, <http://public.dhe.ibm.com/common/ssi/ecm/en/imw14282usen/IMW14282USEN.PDF>



The evolution of database system technology



Source: Data Mining Concepts and Techniques
Jaiwei Han and Macheline Kambler

What is Data Mining?

Data mining consists of the use of algorithms that are directed at one, or more, of four basic areas of exploration. These are:

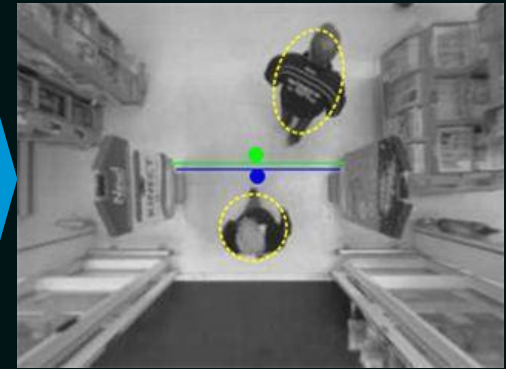
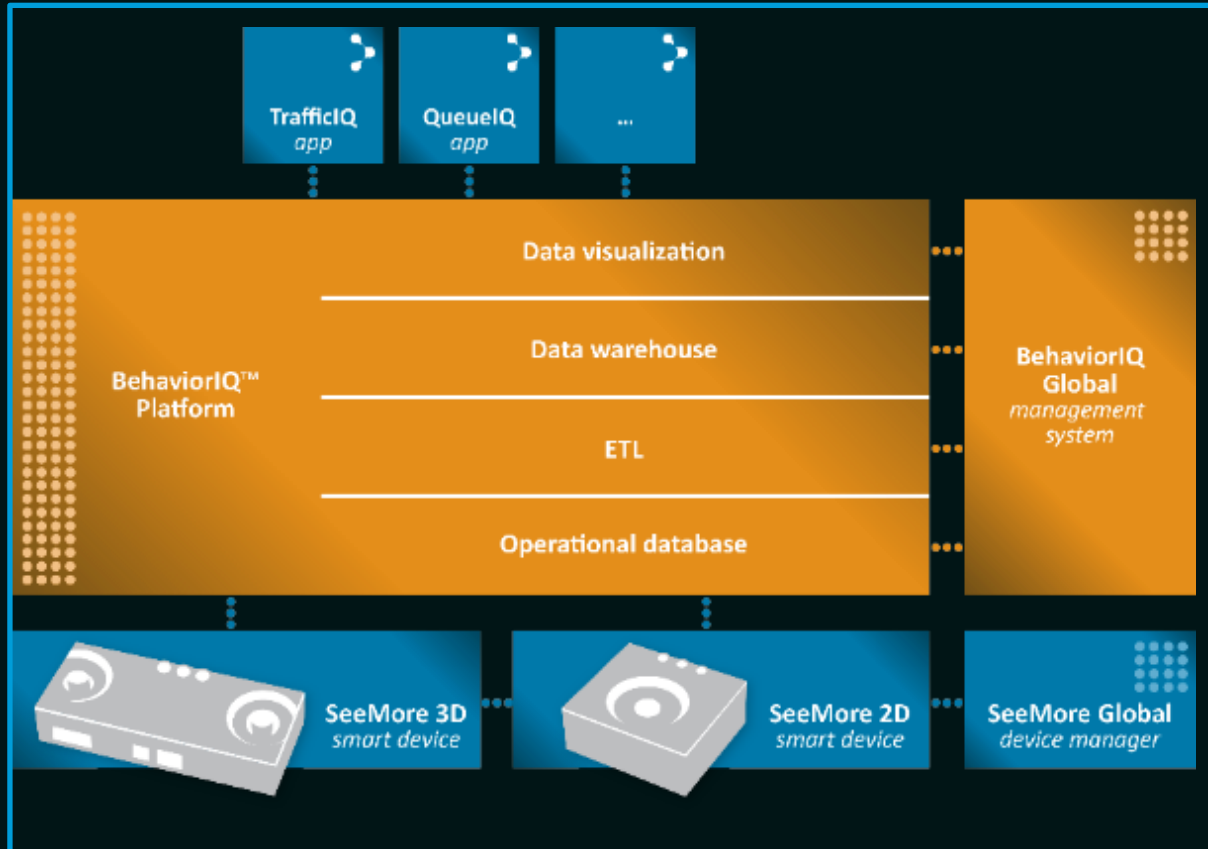
1. **Classification**: the ability to classify a record or case as either belonging to a certain group or of taking on a particular value.
2. **Segmentation**: the ability to create groups (segments or clusters) from a set of data. The groups may be created using either continuous or discrete measures, or a mix of these measures.
3. **Association**: the ability to understand activities that may occur together (for example people who buy bread, also buy milk) or in sequence (for example, people download a song, then download the entire CD)
4. **Anomaly detection**: the ability to find those items, which do not fit the pattern. This is often used in fraud detection or in the identification of aberrant activities.

Source: An Introduction to Data Mining and Predictive Analytics



Classification

Source: <http://www.brickstream.com/solutions/behavior/>

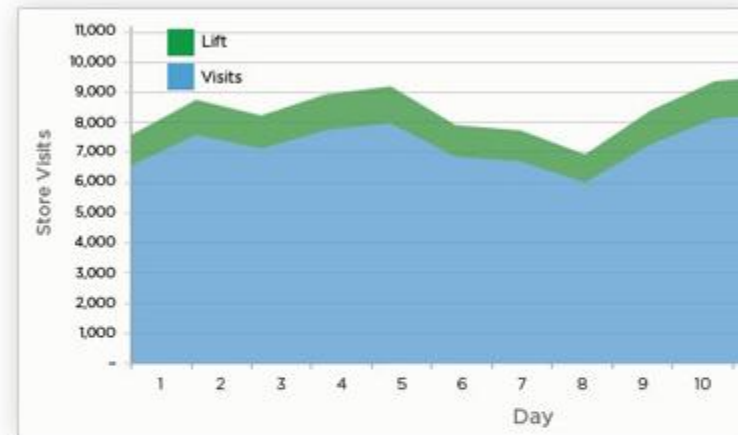


Segmentation

Source: <http://www.placed.com/how-it-works>

Placed is the currency for location with solutions to measure the impact of mobile ads on in-store visits, enable behavioral mobile ad targeting, and deliver real-world consumer insights to drive business decisions.

% TOTAL VISITORS	INDEX	BUSINESS	
51.34%	91		McDonald's View Business Details
44.40%	143		Starbucks View Business Details

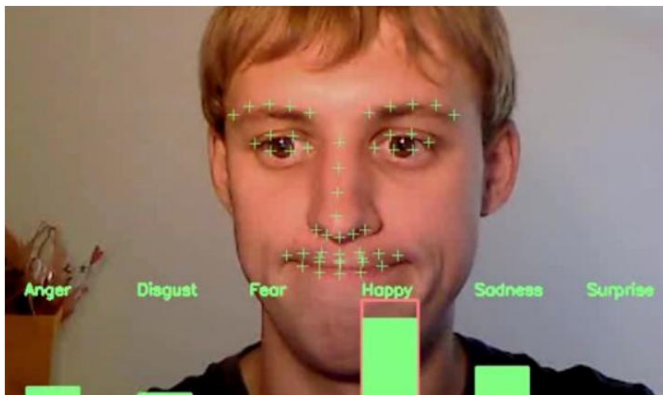


Audience Visualization: Macy's, JCPenney
Market: New York
Source: Placed Targeting, August 2013

 High Affinity Locations for Retailer's Customers

Association & Detection

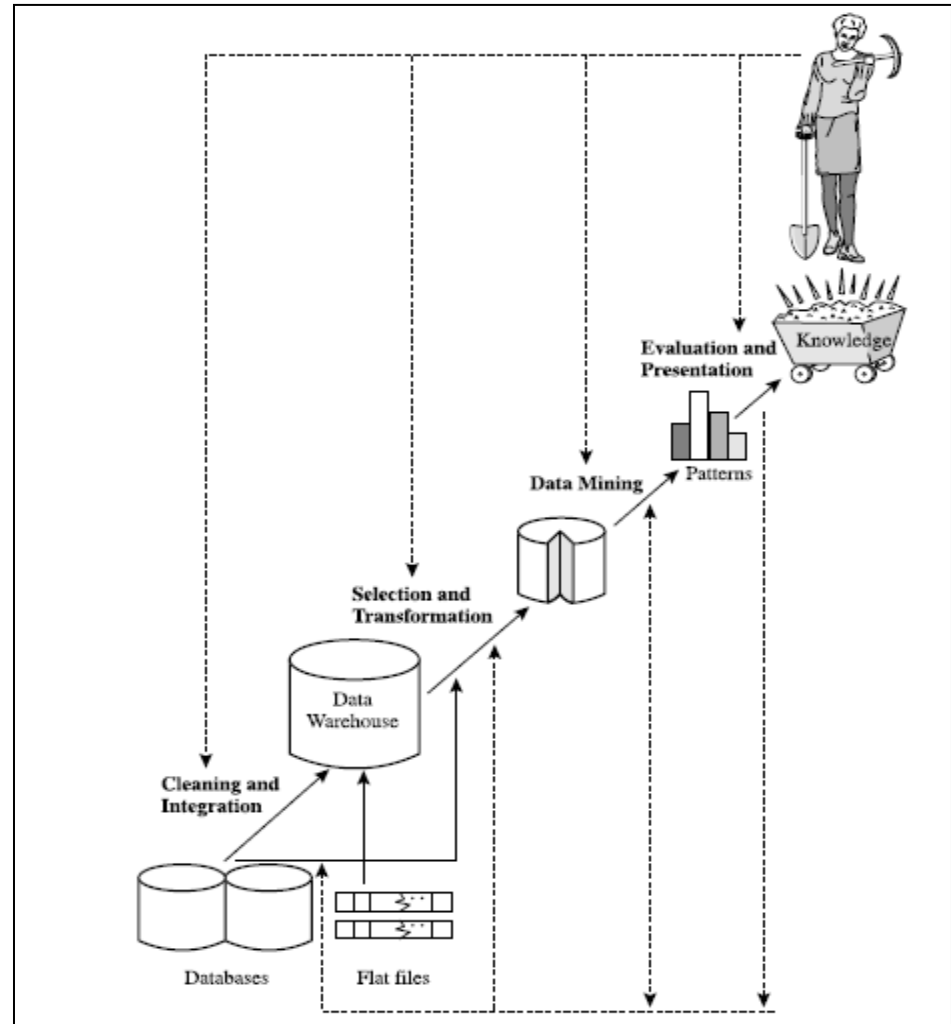
- Realeyes uses computer vision to read people's faces and to measure how they feel.
- Online advertisers can use human reaction and behavior for stronger product branding and improve conversion rate – viewers to buyers.
- Works over regular web cameras and advertisers get results in less than two days.



What is Data Mining?

Data mining, at its core, is the transformation of large amounts of data into meaningful patterns and rules. Further, it could be broken down into two types: directed and undirected. In directed data mining, you are trying to predict a particular data point — the sales price of a house given information about other houses for sale in the neighborhood, for example

In *undirected* data mining, you are trying to create groups of data, or find patterns in existing data — creating the "Soccer Mom" demographic group, for example. In effect, every U.S. census is data mining, as the government looks to gather data about everyone in the country and turn it into useful information.



Source: Data mining with WEKA, Part 1: Introduction and regression <http://www.ibm.com/developerworks/library/os-weka1/>
Data Mining Concepts and Techniques Jaiwei Han and Macheline Kambler

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

What is Data Mining?

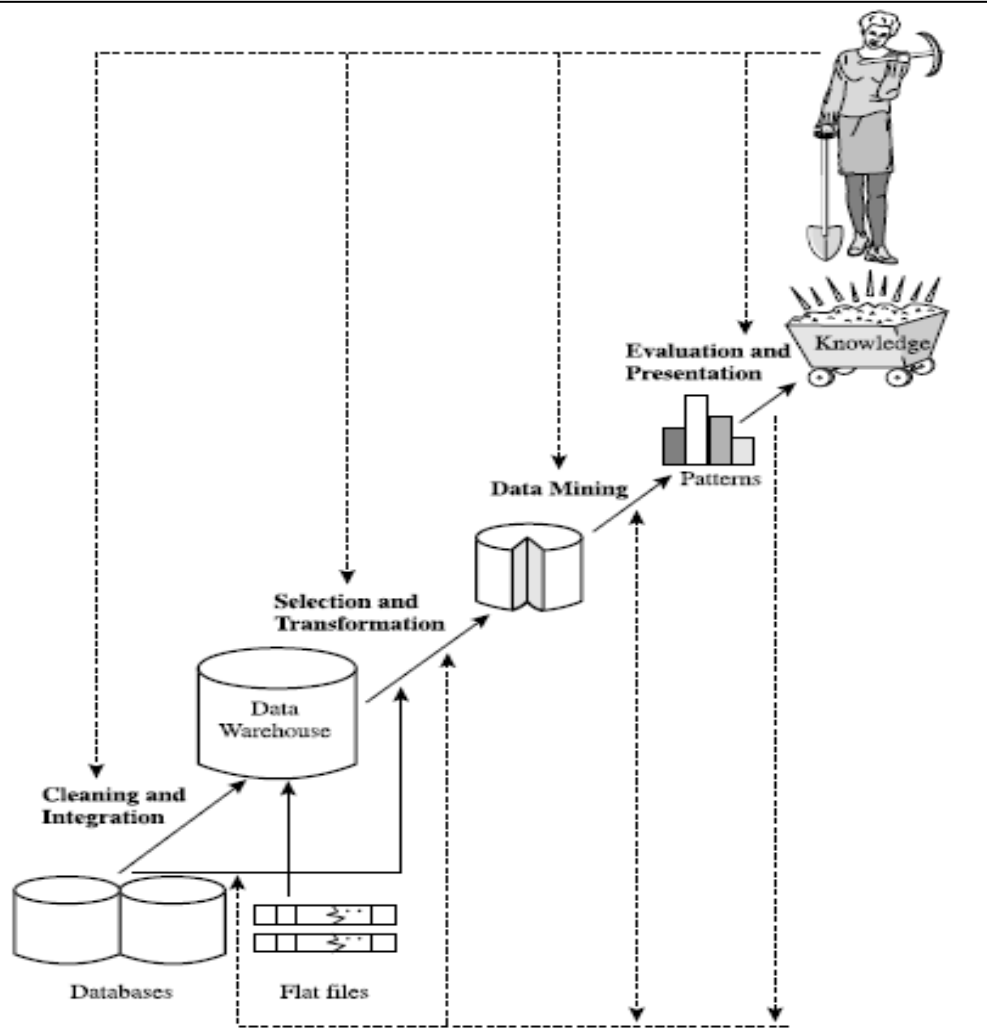
- Data mining (knowledge discovery in databases):
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) information or patterns from data in large databases
- Alternative names and their “inside stories”:
 - Knowledge discovery(mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Data mining involves exploring and analyzing large amounts of data to find patterns for big data. The techniques came out of the fields of statistics and artificial intelligence (AI), with a bit of database management thrown into the mix.
- A common misconception is that data mining involves passing huge amounts of data through intelligent technologies that, alone, find patterns and give magical solutions to business problems. This is not true, although there is more automation than in traditional statistical applications.
- **Data mining is an interactive and iterative process. *Business expertise must be used jointly with advanced technologies to identify underlying relationships and features in the data.*** A seemingly useless pattern in data discovered by data-mining technology can often be transformed into a valuable piece of actionable information using business experience and expertise.

Data mining is also sometimes called Knowledge Discovery in Databases (KDD).



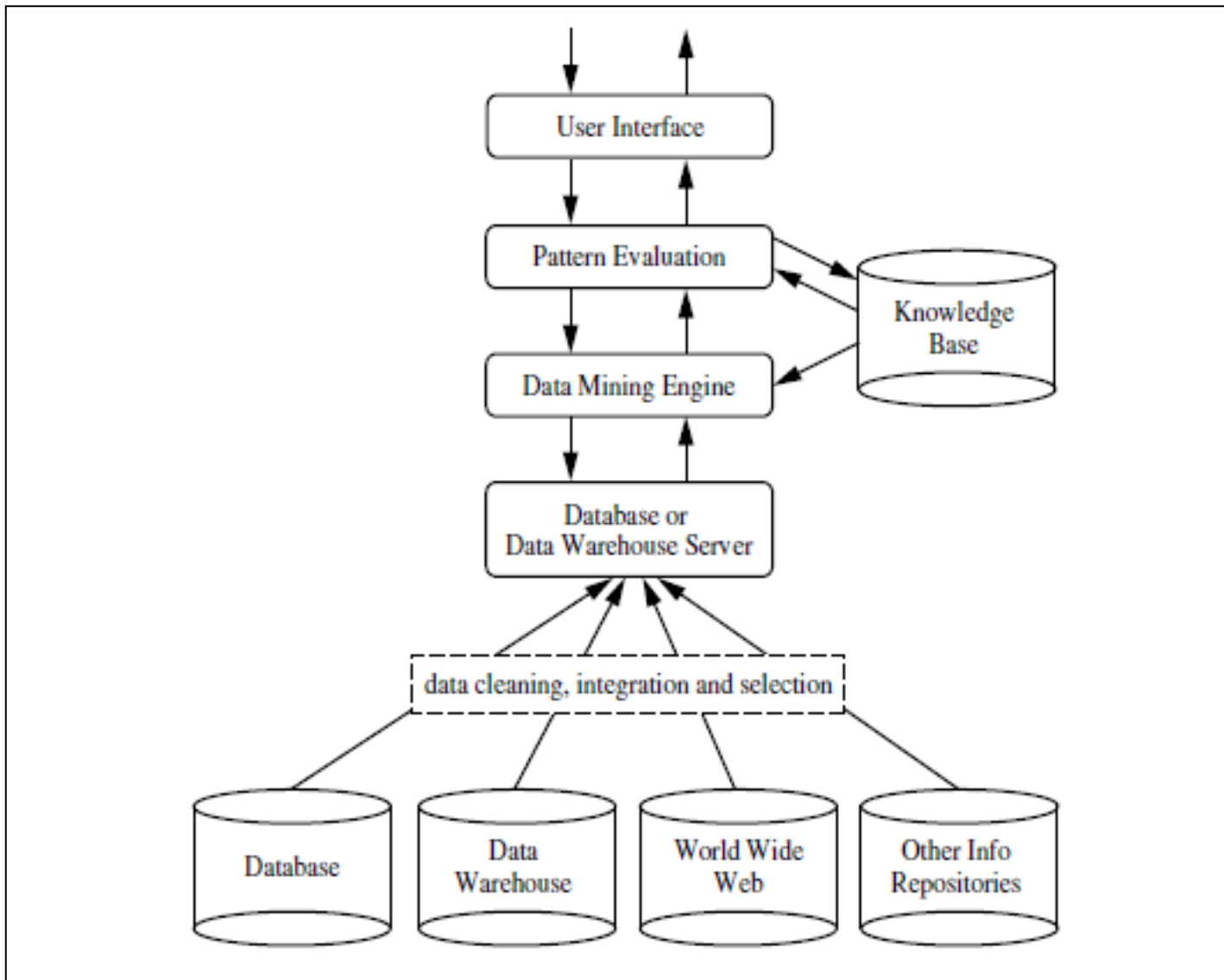
Knowledge Discovery Process

Data Mining as a step in the process of Knowledge discovery.



1. **Data Cleansing** (to remove noise and inconsistent data)
2. **Data Integration** (where multiple data sources may be combined)
3. **Data selection** (where data relevant to the analysis task are retrieved from the database)
4. **Data Transformation** (where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations, for instance)
5. **Data Mining** (an essential process where intelligent methods are applied in order to extract data patterns)
6. **Pattern evaluation** (to identify the truly interesting patterns representing knowledge based on some interestingness measures)
7. **Knowledge presentation** (where visualization and knowledge representation techniques are used to present the mined knowledge to the user)

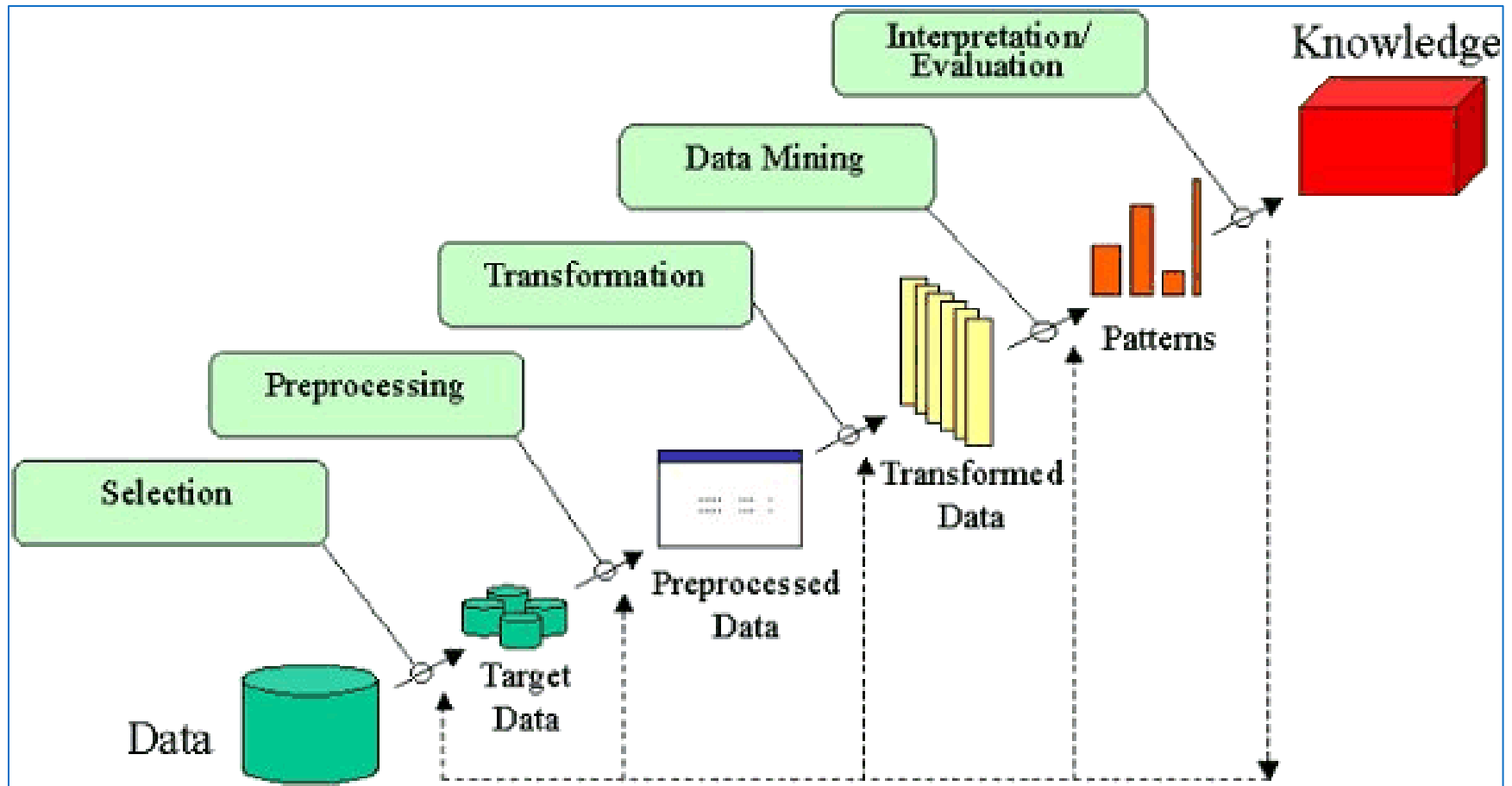
Architecture of a Data Mining System



Source: Data Mining Concepts and Techniques
Jaiwei Han and Macheline Kamber

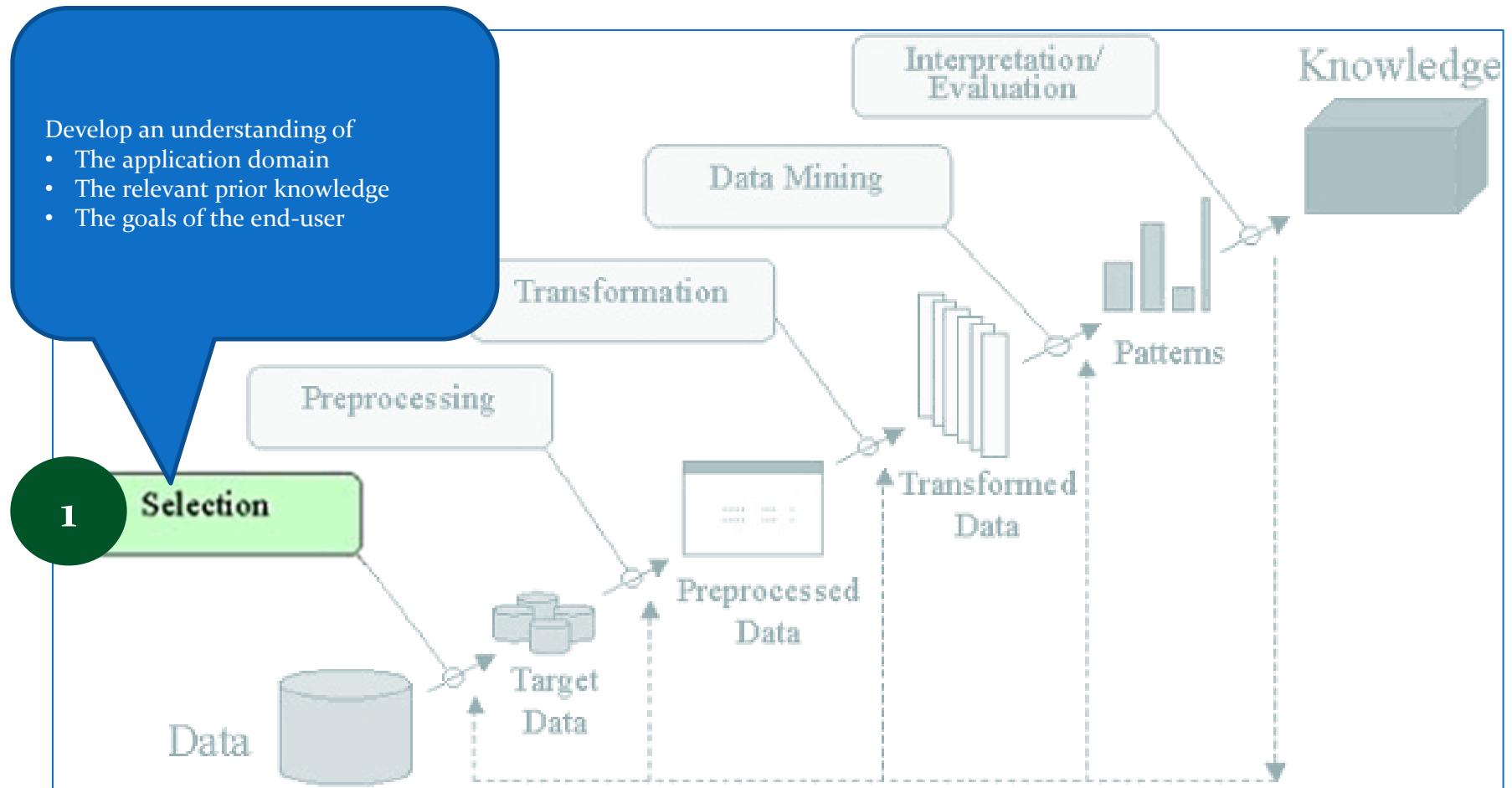


Knowledge Discovery in Databases (KDD)



Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

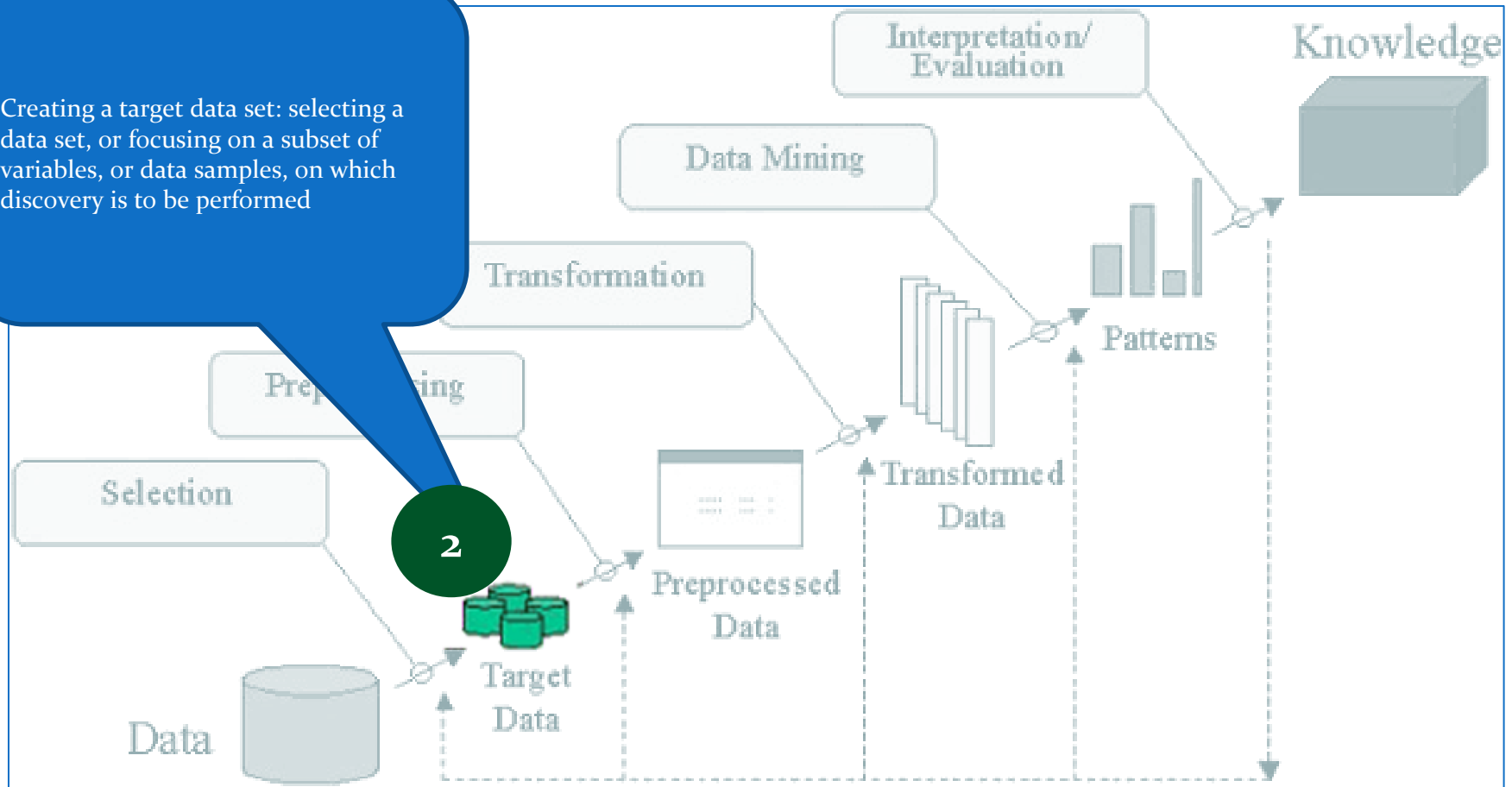
Knowledge Discovery in Databases (KDD)



Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)

Creating a target data set: selecting a data set, or focusing on a subset of variables, or data samples, on which discovery is to be performed



Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)

Data cleaning and preprocessing.

- Removal of noise or outliers.
- Collecting necessary information to model or account for noise.
- Strategies for handling missing data fields.
- Accounting for time sequence information and known changes.

3

Preprocessing

Selection

Data

Transformation

Data Mining

Interpretation/
Evaluation

Knowledge

Target
Data

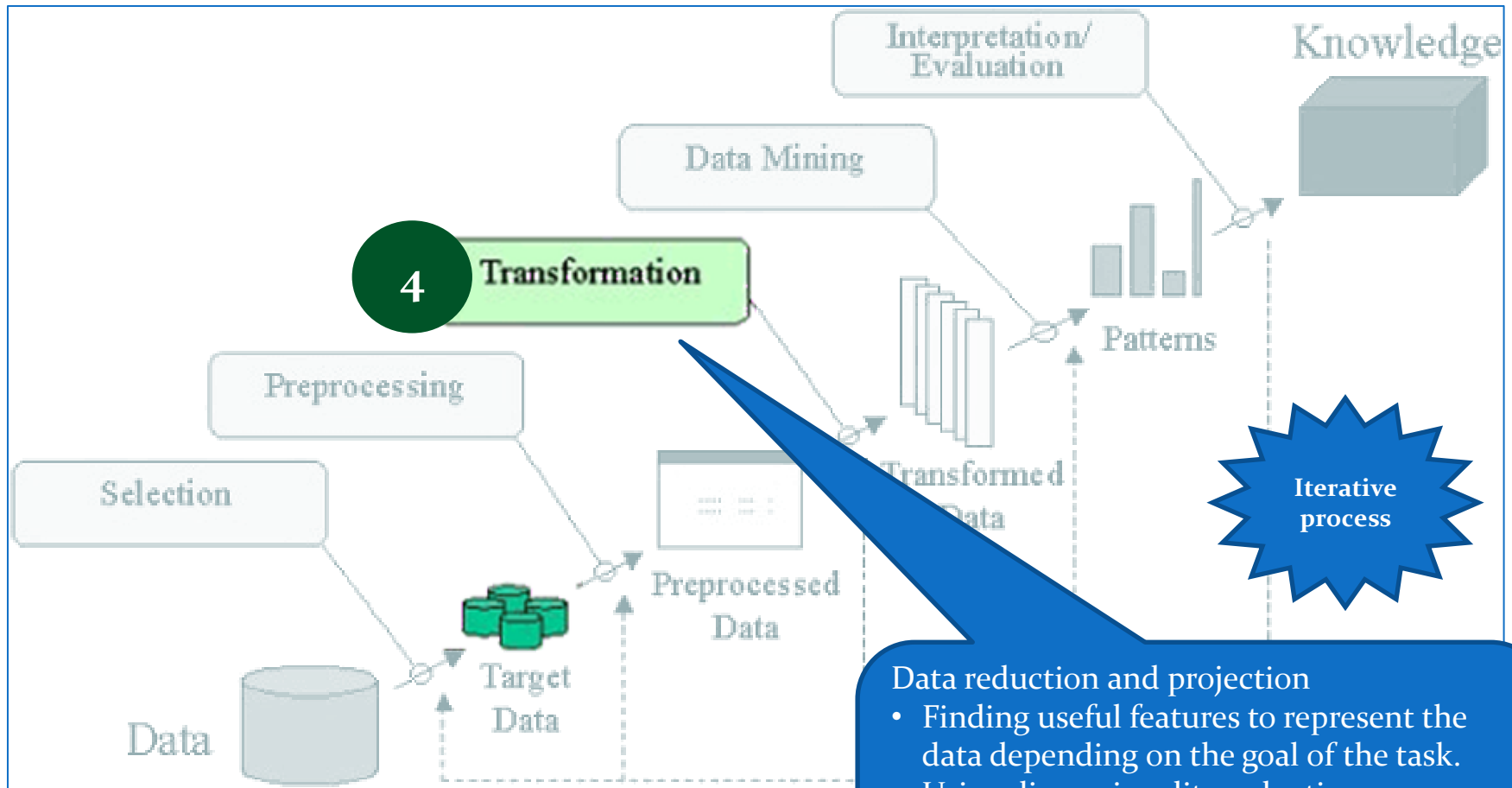
Preprocessed
Data

Transformed
Data

Patterns

Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)



Data reduction and projection

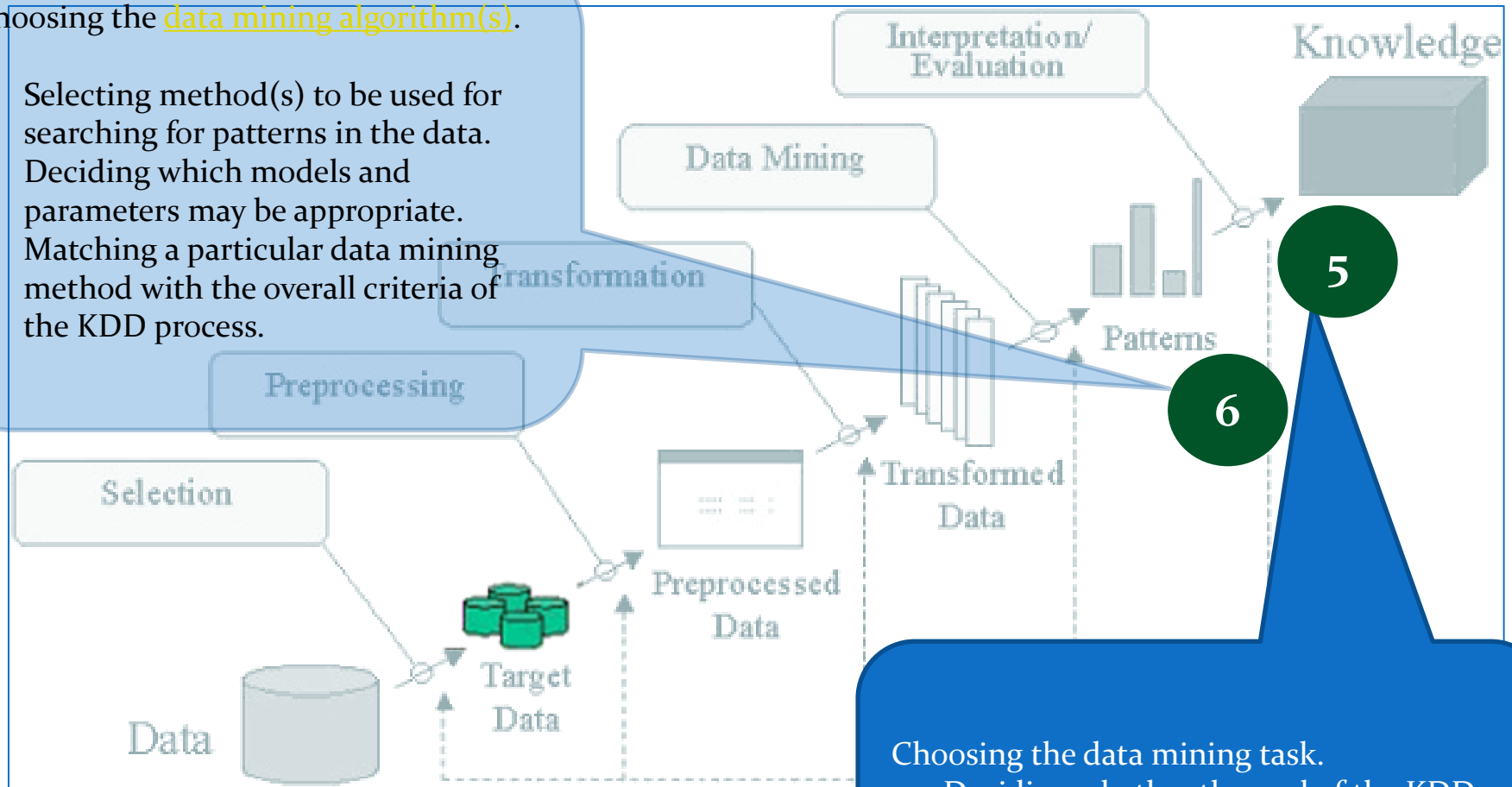
- Finding useful features to represent the data depending on the goal of the task.
- Using dimensionality reduction or transformation methods to reduce the effective number of variables under consideration or to find invariant representations for the data.

Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.htm

Knowledge Discovery in Databases (KDD)

Choosing the data mining algorithm(s).

- Selecting method(s) to be used for searching for patterns in the data.
- Deciding which models and parameters may be appropriate.
- Matching a particular data mining method with the overall criteria of the KDD process.

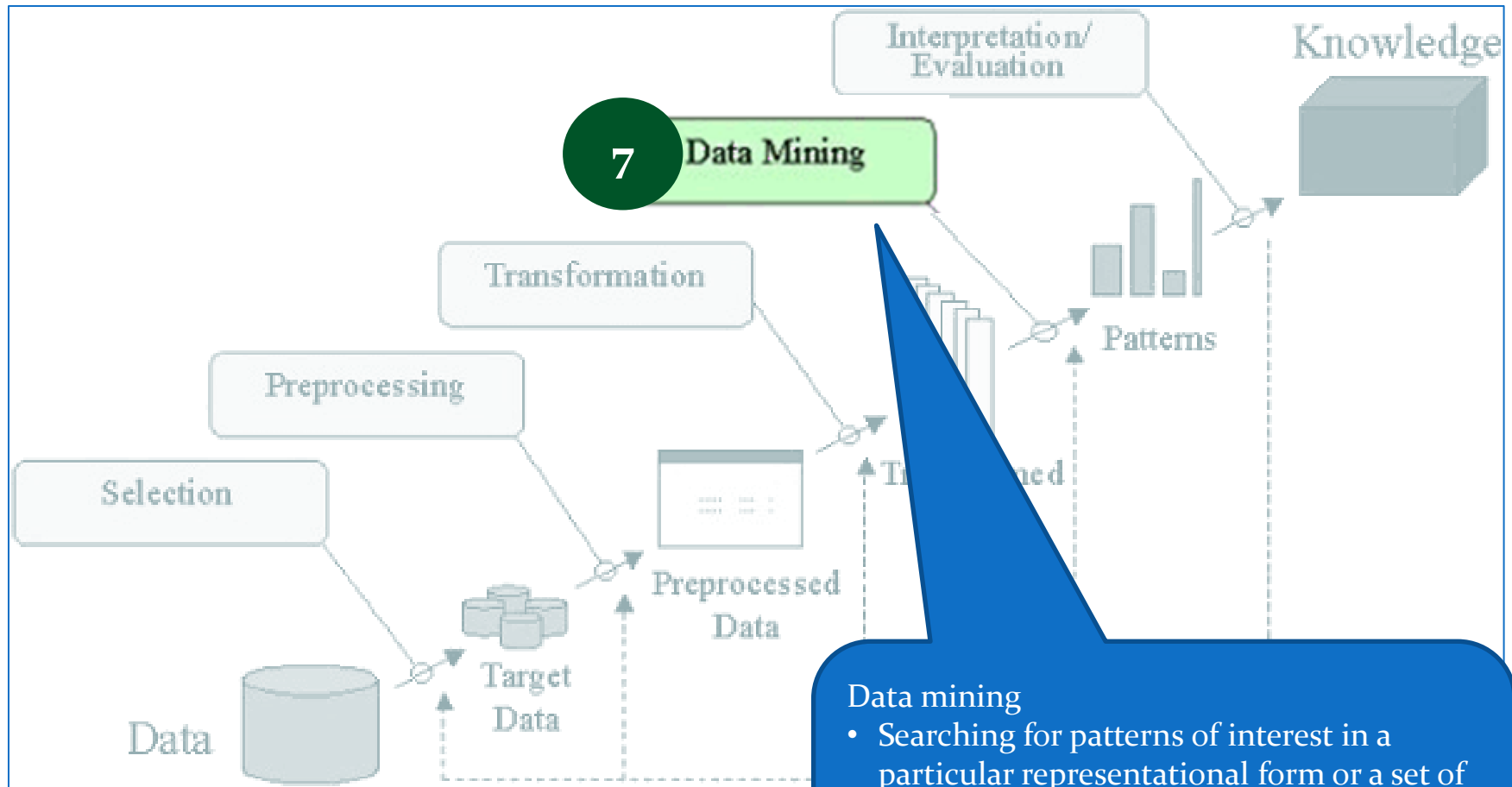


Choosing the data mining task.

- Deciding whether the goal of the KDD process is classification, regression, clustering, etc.

Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)

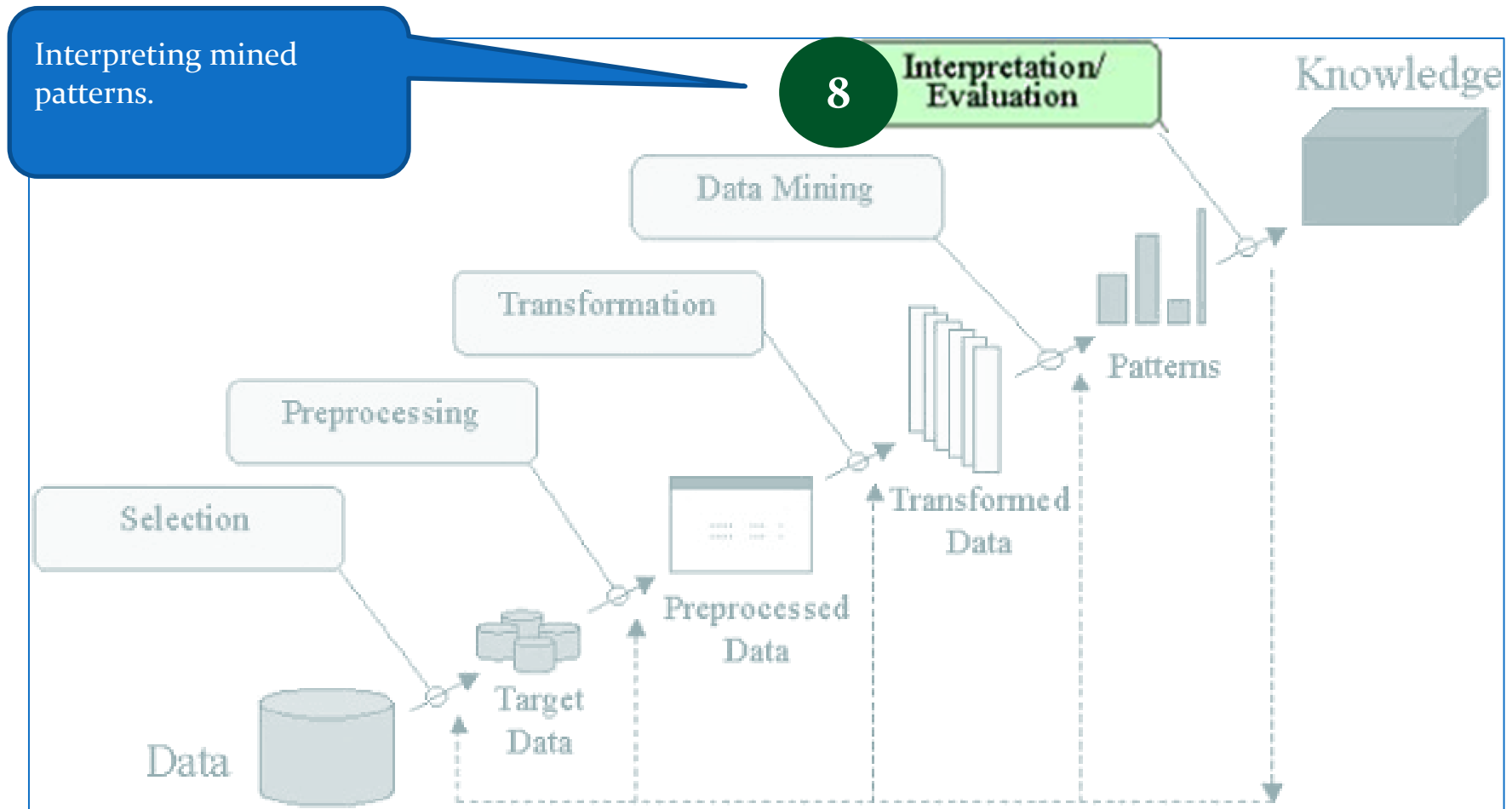


Data mining

- Searching for patterns of interest in a particular representational form or a set of such representations as classification rules or trees, regression, clustering, and so forth.

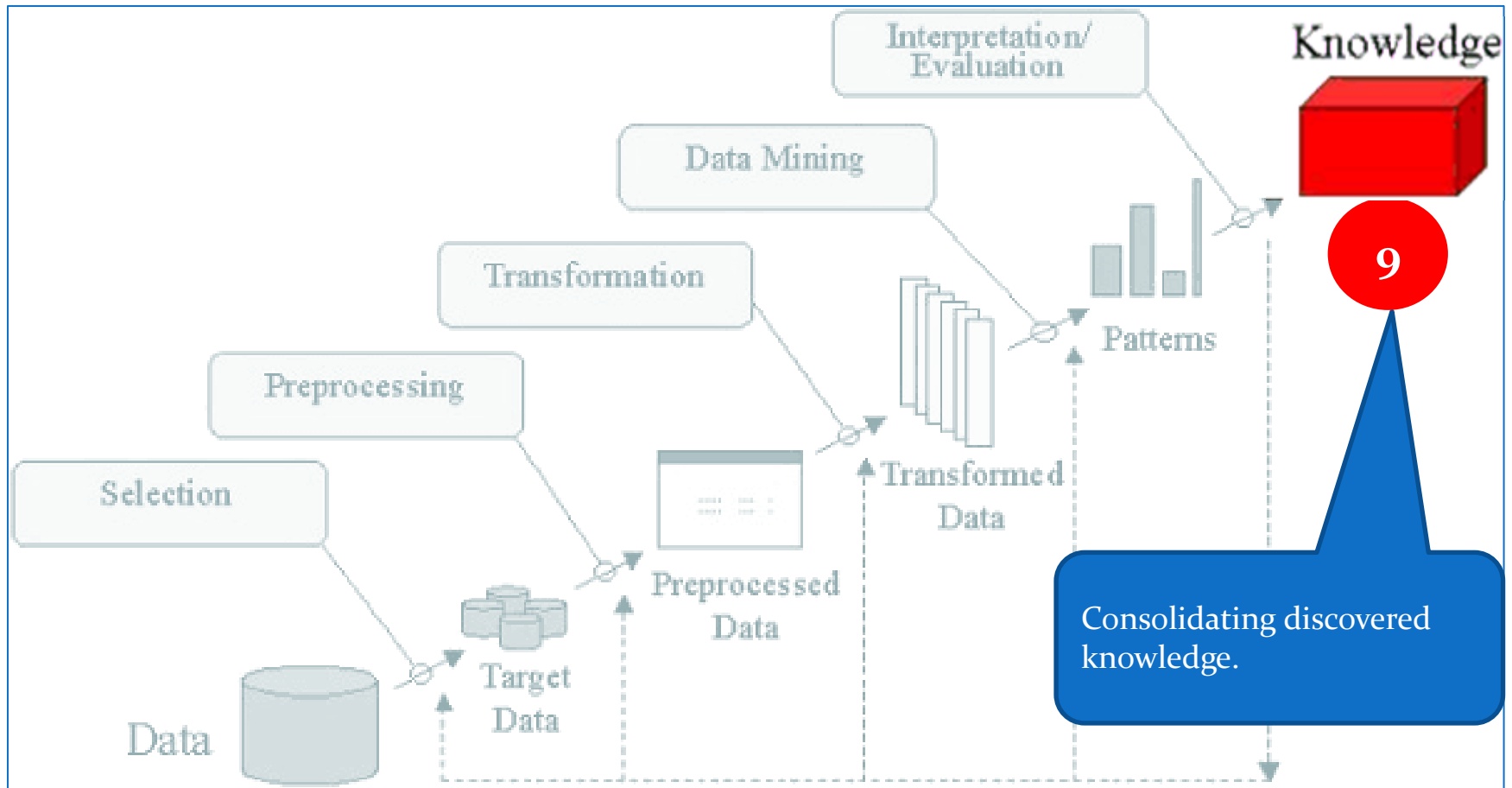
Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)



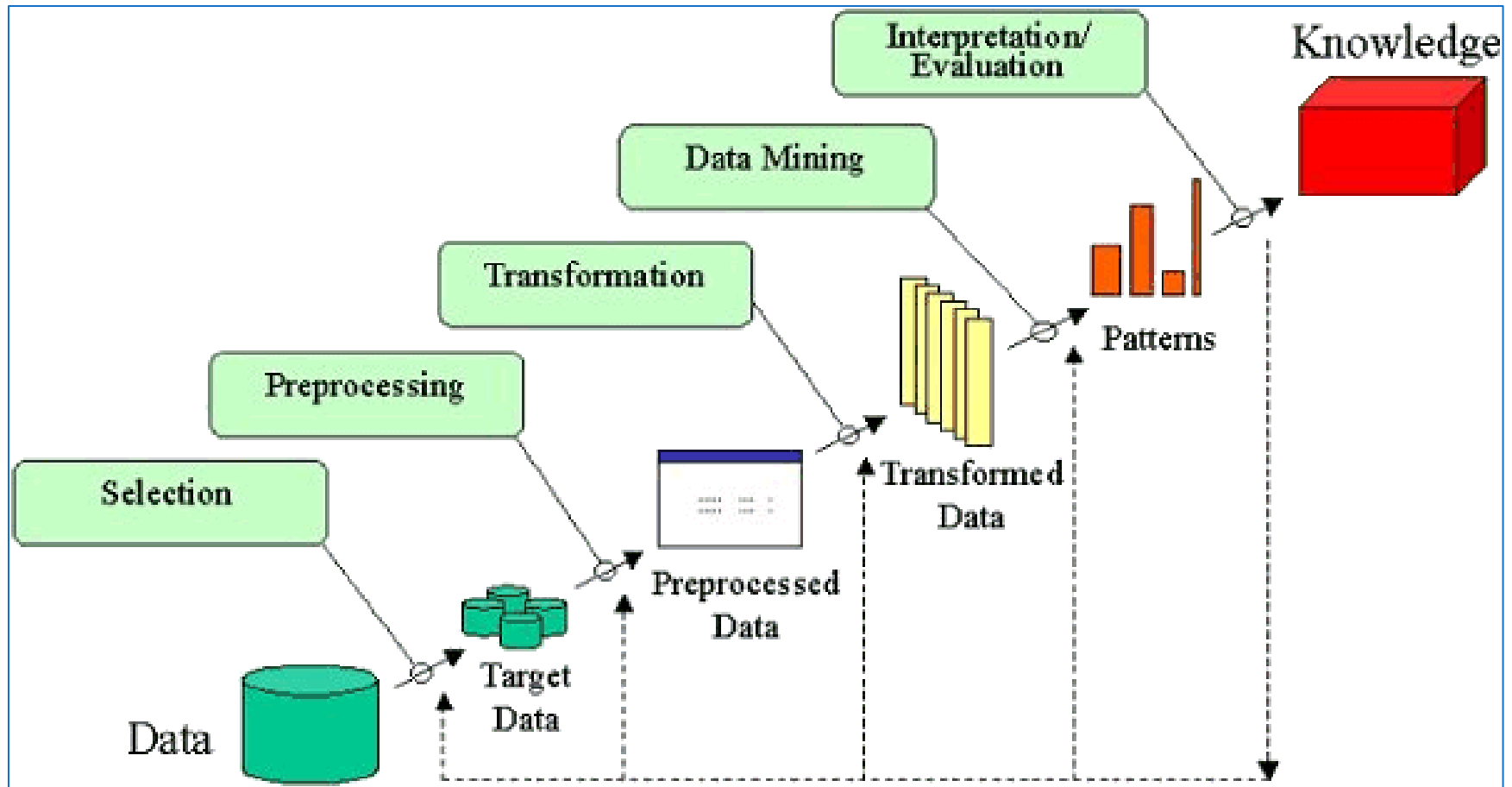
Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)



Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html

Knowledge Discovery in Databases (KDD)



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Knowledge Discovery in Databases (KDD)

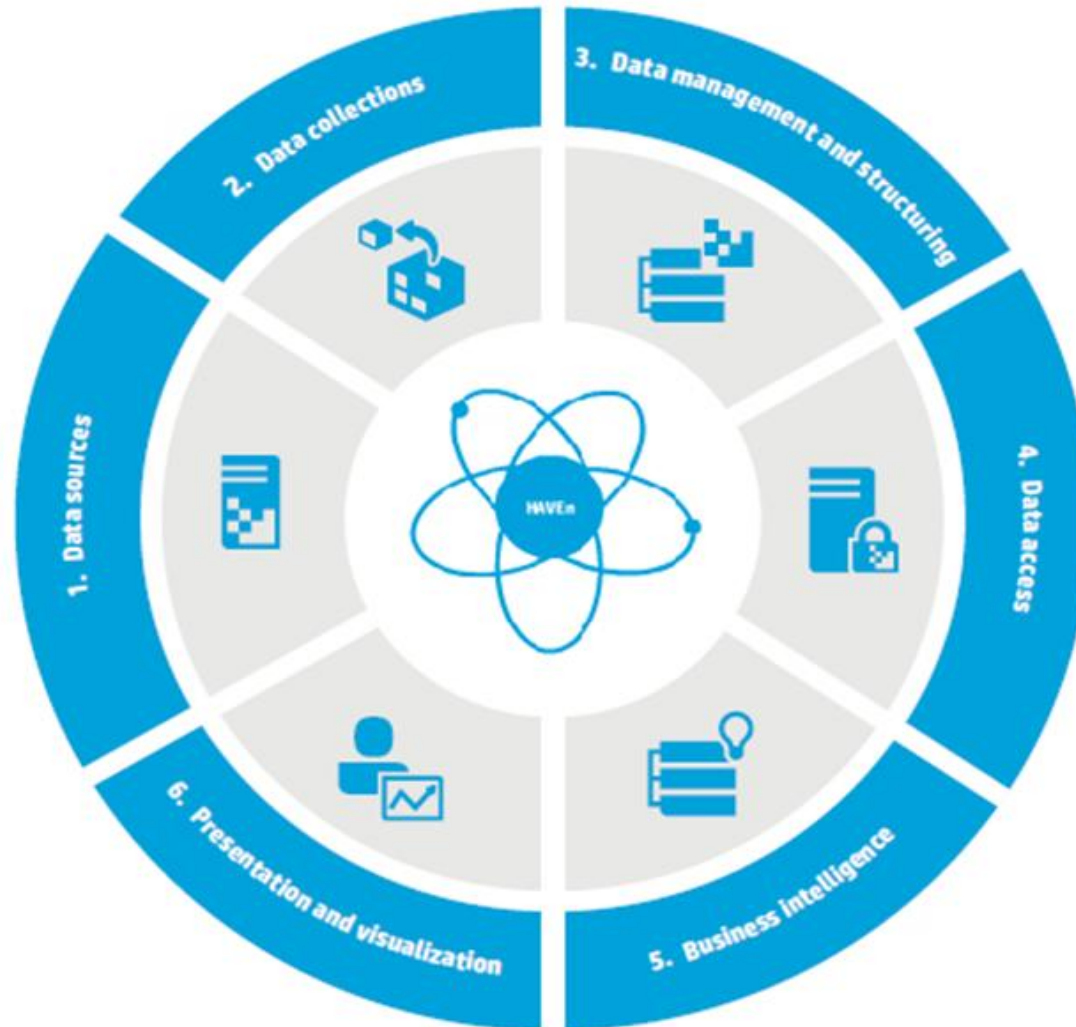
The overall process of finding and interpreting patterns from data involves the repeated application of the following steps:

1. Developing an understanding of
 1. the application domain
 2. the relevant prior knowledge
 3. the goals of the end-user
2. Creating a target data set: selecting a data set, or focusing on a subset of variables, or data samples, on which discovery is to be performed.
3. Data cleaning and preprocessing.
 1. Removal of noise or outliers.
 2. Collecting necessary information to model or account for noise.
 3. Strategies for handling missing data fields.
 4. Accounting for time sequence information and known changes.
4. Data reduction and projection.
 1. Finding useful features to represent the data depending on the goal of the task.
 2. Using dimensionality reduction or transformation methods to reduce the effective number of variables under consideration or to find invariant representations for the data.
5. Choosing the [data mining task](#).
 1. Deciding whether the goal of the KDD process is classification, regression, clustering, etc.
6. Choosing the [data mining algorithm\(s\)](#).
 1. Selecting method(s) to be used for searching for patterns in the data.
 2. Deciding which models and parameters may be appropriate.
 3. Matching a particular data mining method with the overall criteria of the KDD process.
7. Data mining.
 1. Searching for patterns of interest in a particular representational form or a set of such representations as classification rules or trees, regression, clustering, and so forth.
8. Interpreting mined patterns.
9. Consolidating discovered knowledge.

Source: Overview of the KDD process - http://www2.cs.uregina.ca/~dbd/cs831/notes/kdd/1_kdd.html



HP HAVEn



Source: HP HAVEn

Key Questions for a Data-Mining Project

Before considering which specific data-mining technique is suitable, the business problem and the data need to be assessed for any potential data-mining project. There are several aspects which should be considered:

- Are data available?
- Do data cover the relevant factors?
- Are the data too noisy?
- Are there enough data?
- Is expertise on the data available?
 - Domain experts...

Source: An Introduction to Data Mining and Predictive Analytics



Strategy for Data Mining: the CRISP-DM

Process Methodology

The majority work in data mining requires the careful eye of a knowledgeable business analyst to keep the process on track. To guide your planning, answer the following questions: :

- What substantive problem do you want to solve?
- What data sources are available, and what parts of the data are relevant to the current problem?
- What kind of preprocessing and data cleaning do you need to do before you start mining the data?
- What data mining technique(s) will you use?
- How will you evaluate the results of the data mining analysis?
- How will you get the most out of the information you obtained from data mining?

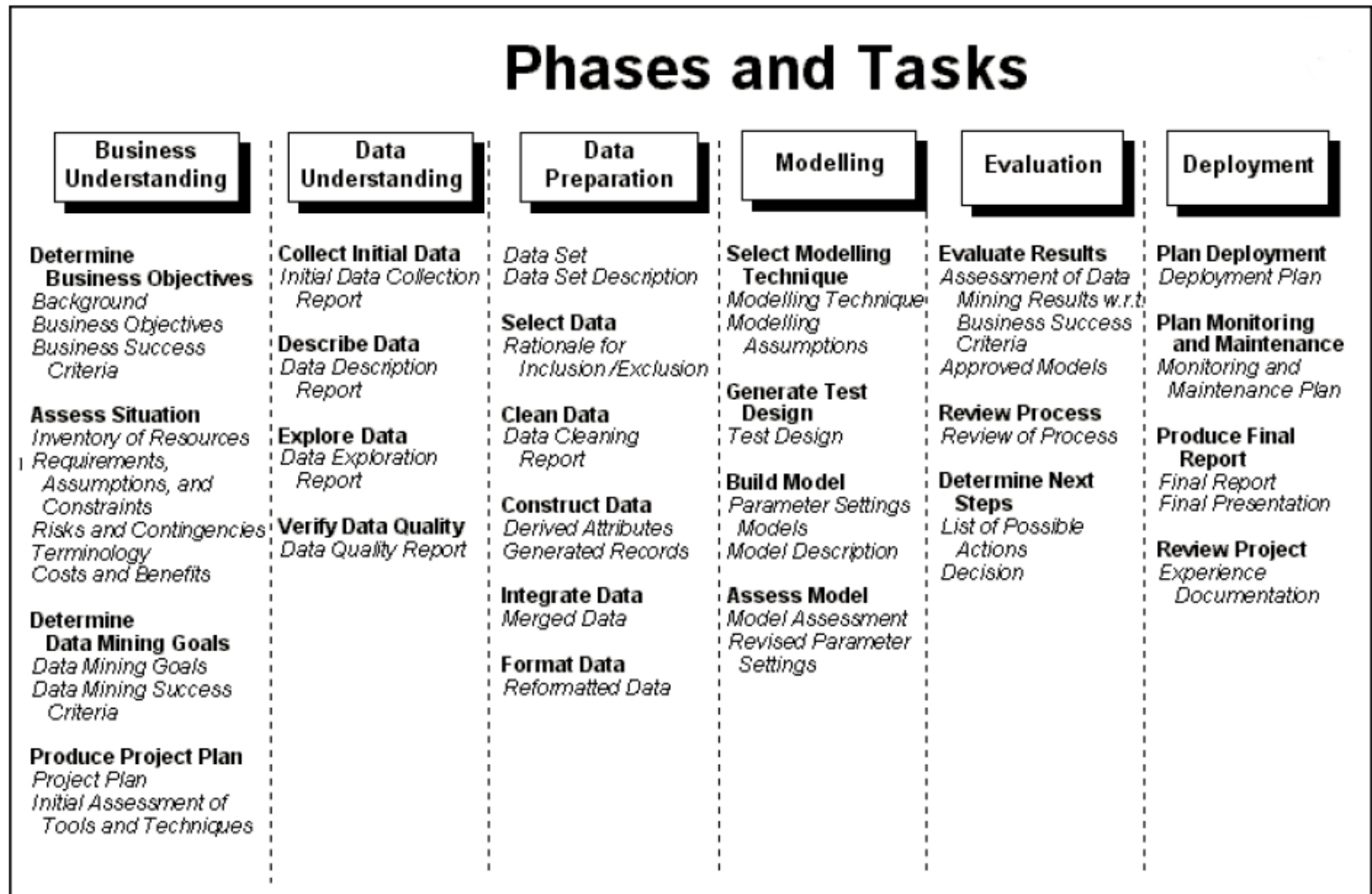
The typical data-mining process can become complicated very quickly. There is a lot to keep track of—complex business problems, multiple data sources, varying data quality across data sources, an array of data mining techniques, different ways of measuring data mining success, and so on.

To stay on track, it helps to have an explicitly defined process model for data mining. The process model guides you through the critical issues outlined above and makes sure that the important points are addressed. It serves as a data mining road map so that you won't lose your way as you dig into the complexities of your data.



The CRISP-DM Process Methodology

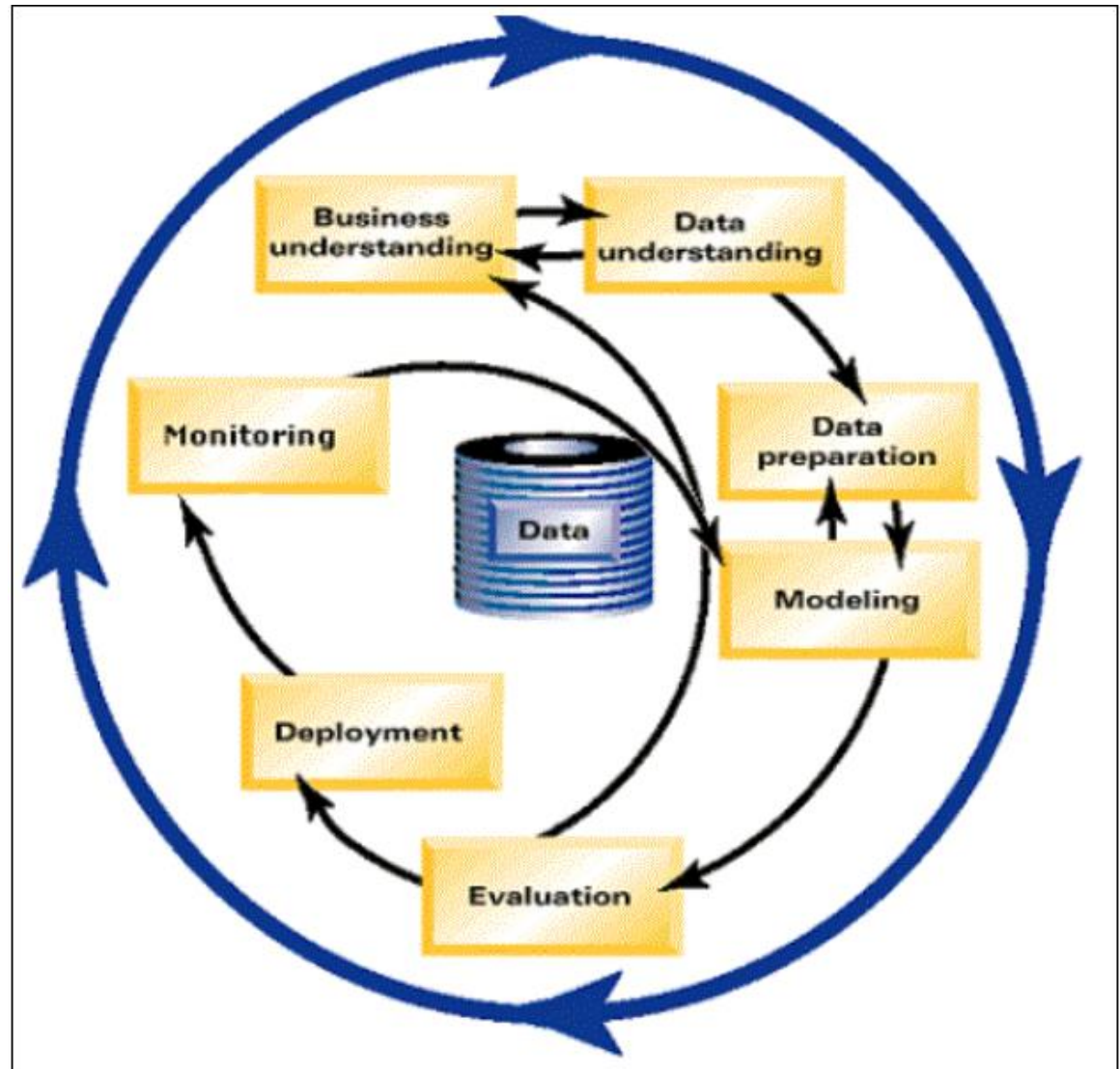
The Cross-Industry Standard Process for Data Mining (CRISP-DM) is the model that is designed as a general model that can be applied to a wide variety of industries and business problems.



Source: Introduction
Modeler and Data Mi
[ftp://ftp.software.ibm.com/analytics/spss/support/mentation/14/UserMa](ftp://ftp.software.ibm.com/analytics/spss/support/mentation/14/UserManual/CRISP-DM.pdf)
[DM.pdf](ftp://ftp.software.ibm.com/analytics/spss/support/mentation/14/UserManual/CRISP-DM.pdf)

Stages in CRISP-DM

The figure below illustrates the main stages in a successful data mining process: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation and Deployment (we have also added monitoring of model performance, which as just noted is crucial). The outer circle represents the fact that the whole process is iterative. The clockwise order of the tasks represents the common sequence.



Source: Introduction to IBM SPSS Modeler and Data Mining,
[ftp://ftp.software.ibm.com/software/analytics/spss/support/Modeler/Documentation/14/Use
rManual/CRISP-DM.pdf](ftp://ftp.software.ibm.com/software/analytics/spss/support/Modeler/Documentation/14/Use%20Manual/CRISP-DM.pdf)

THINK

BIG

BIG

Data Mining – Kind of Data

In principle, data mining should be applicable to any kind of data repository, as well as to transient data, such as data streams. Thus the scope of our examination of data repositories will include relational databases, data warehouses, transactional databases, advanced database systems, flat files, data streams, and the World Wide Web. Advanced database systems include object-relational databases and specific application-oriented databases, such as spatial databases, time-series databases, text databases, and multimedia databases. The challenges and techniques of mining may differ for each of the repository systems.

- Relational Databases
- Data Warehouses
- Transactional Databases
- Object Relational Databases
- Temporal Databases, Sequence Databases, and Time-Series Databases
- Spatial Databases and Spatiotemporal Databases
- Text Databases and Multimedia Databases
- Heterogeneous Databases and Legacy Databases
- Data Streams
- The World Wide Web

Source: An Introduction to Data Mining and Predictive Analytics



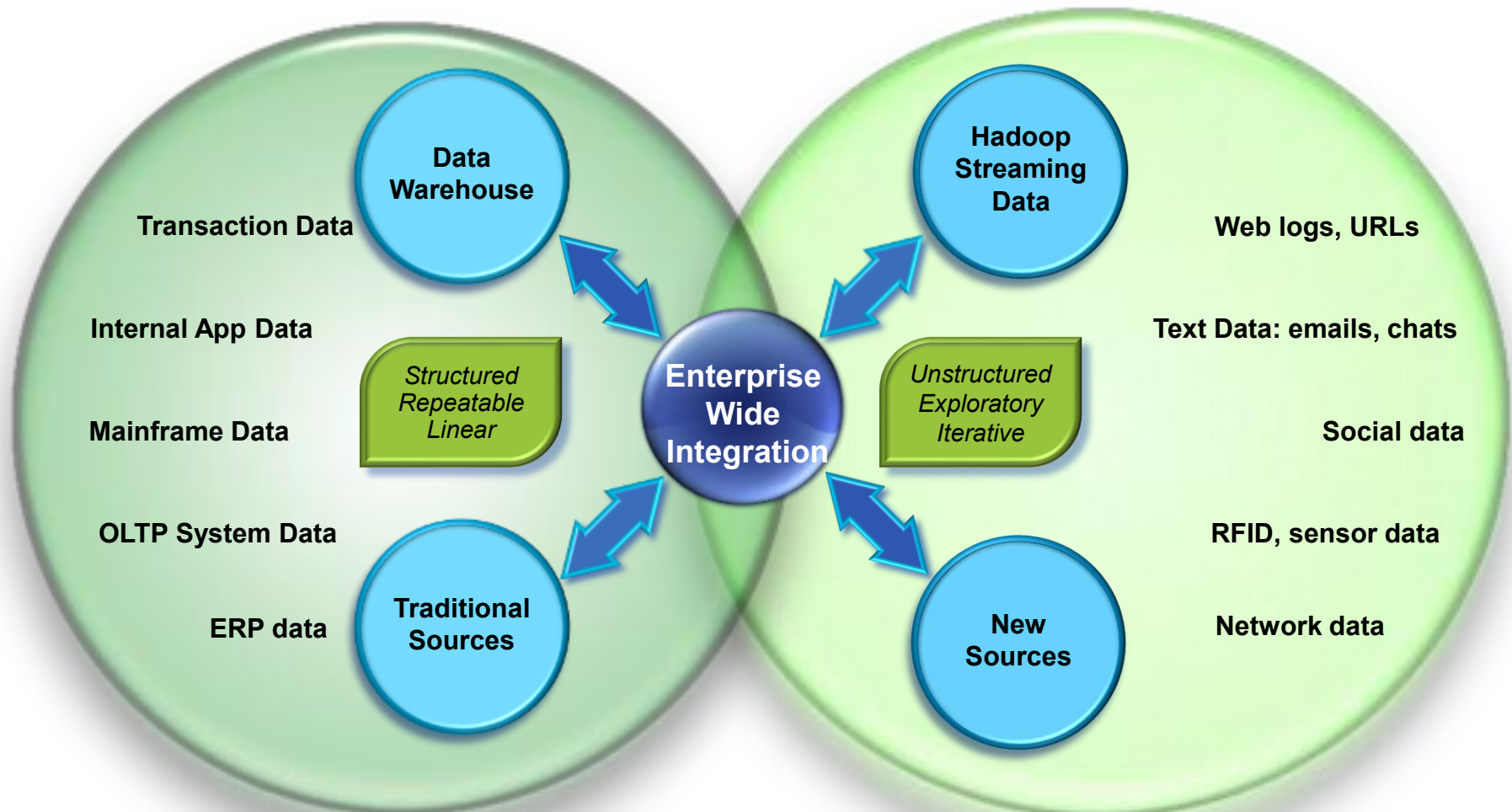
Analysis expanding from enterprise data to big data, creating new cost-effective opportunities for competitive advantage

Traditional Approach

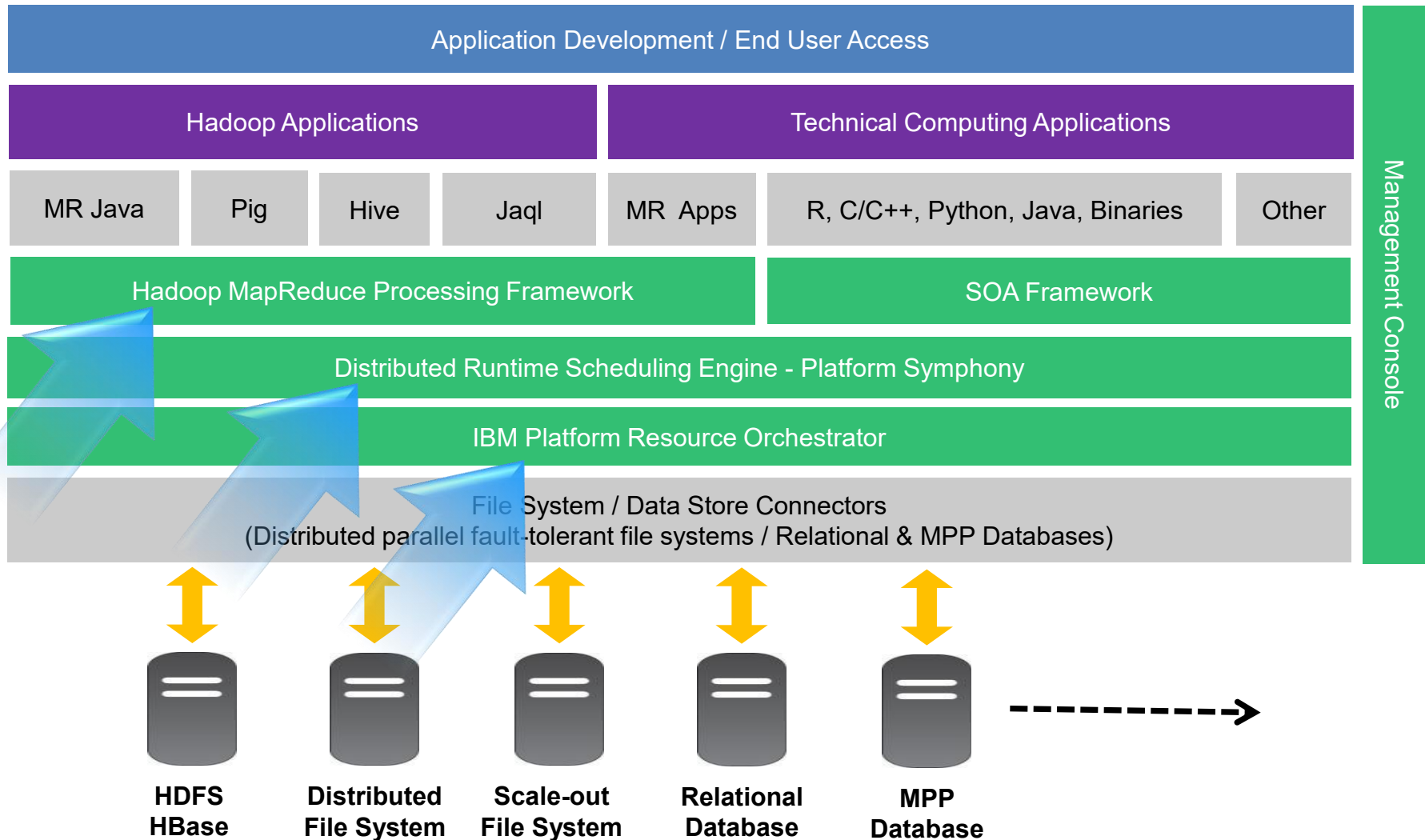
Structured, analytical, logical

New Approach

Creative, holistic thought, intuition



Application & Data Integration Architecture



Data Mining – Kind of Data

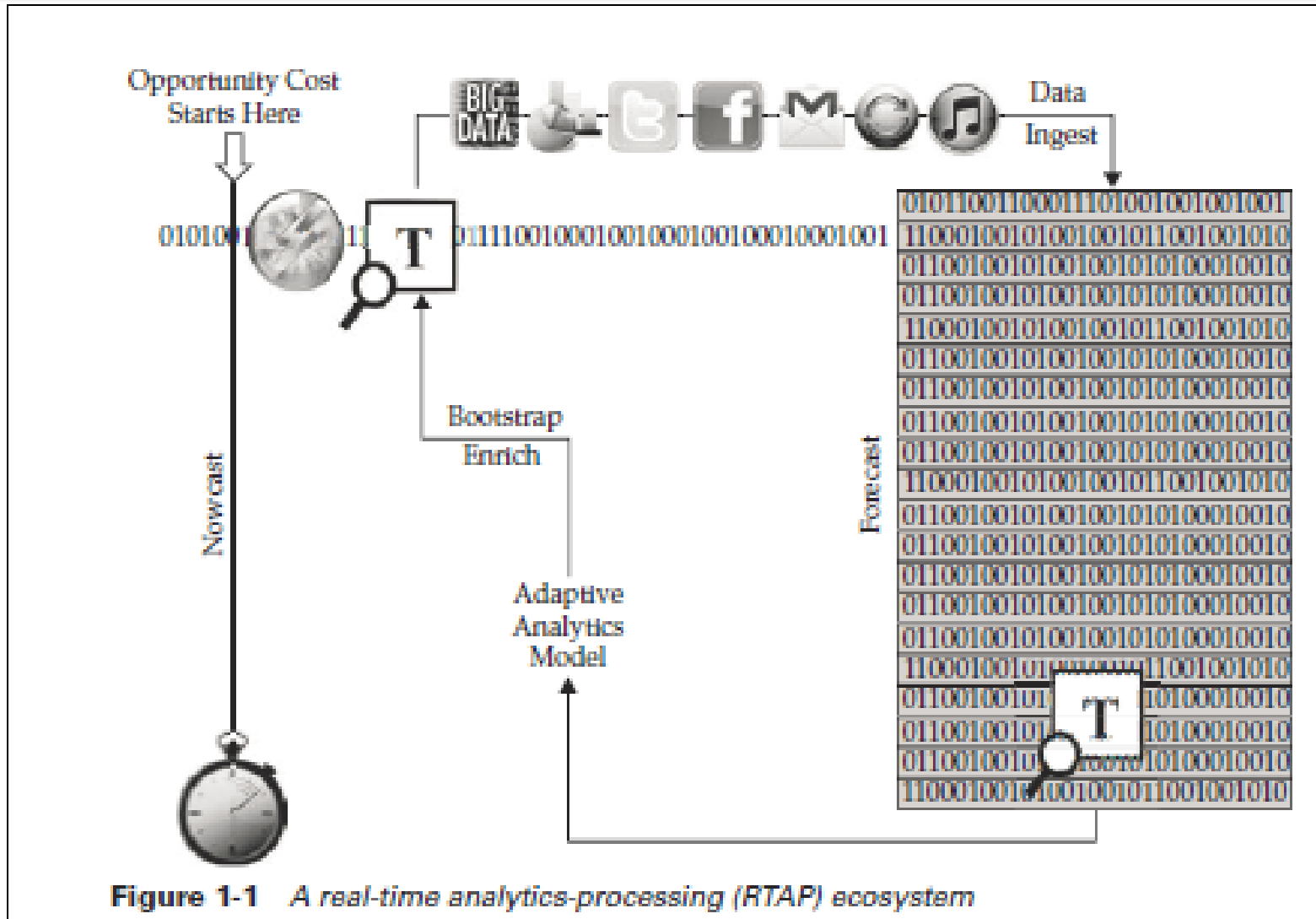
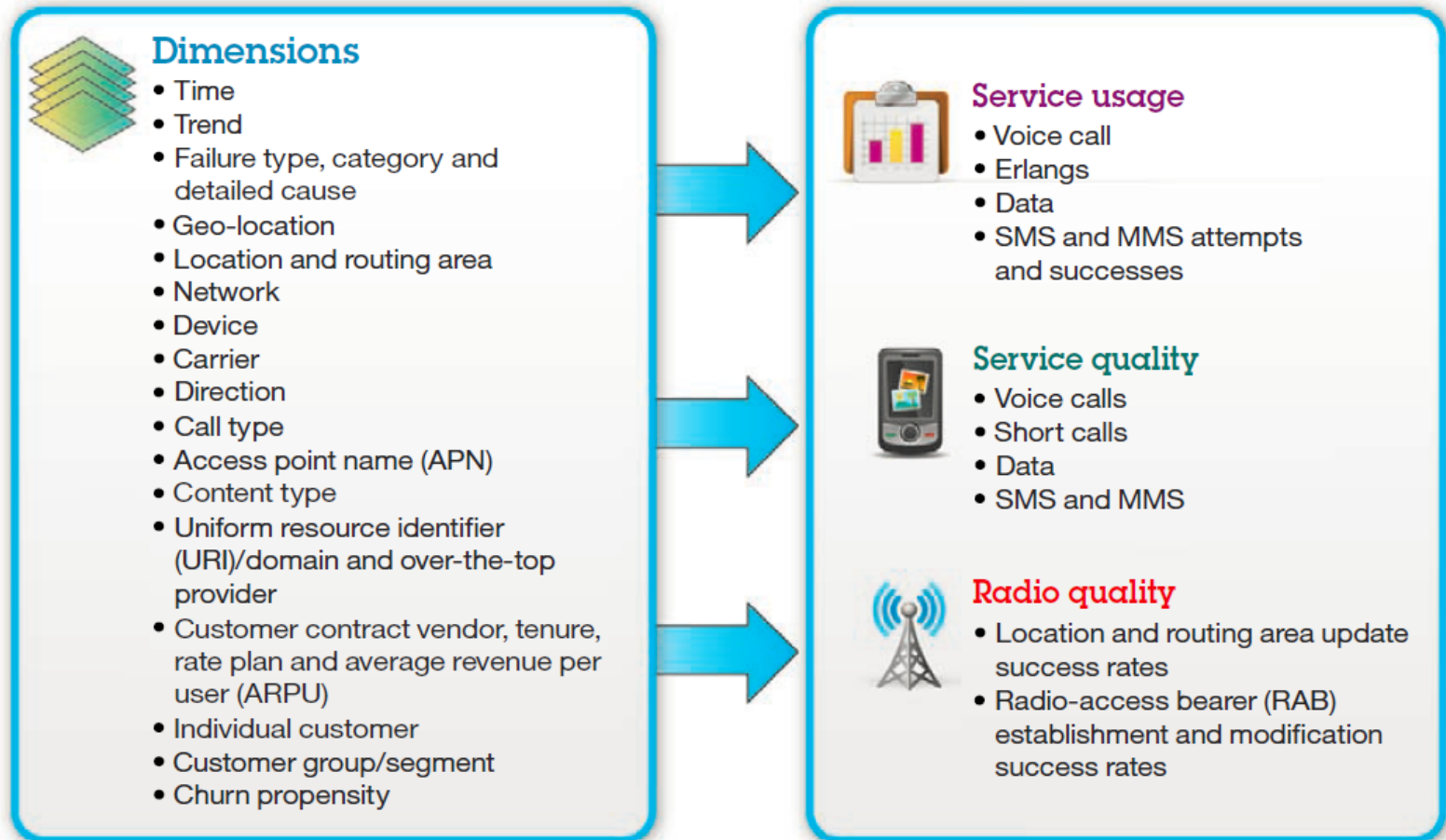


Figure 1-1 A real-time analytics-processing (RTAP) ecosystem

Source: Harness the power of Big data - <http://public.dhe.ibm.com/common/ssi/ecm/en/imm14100usen/IMM14100USEN.PDF>

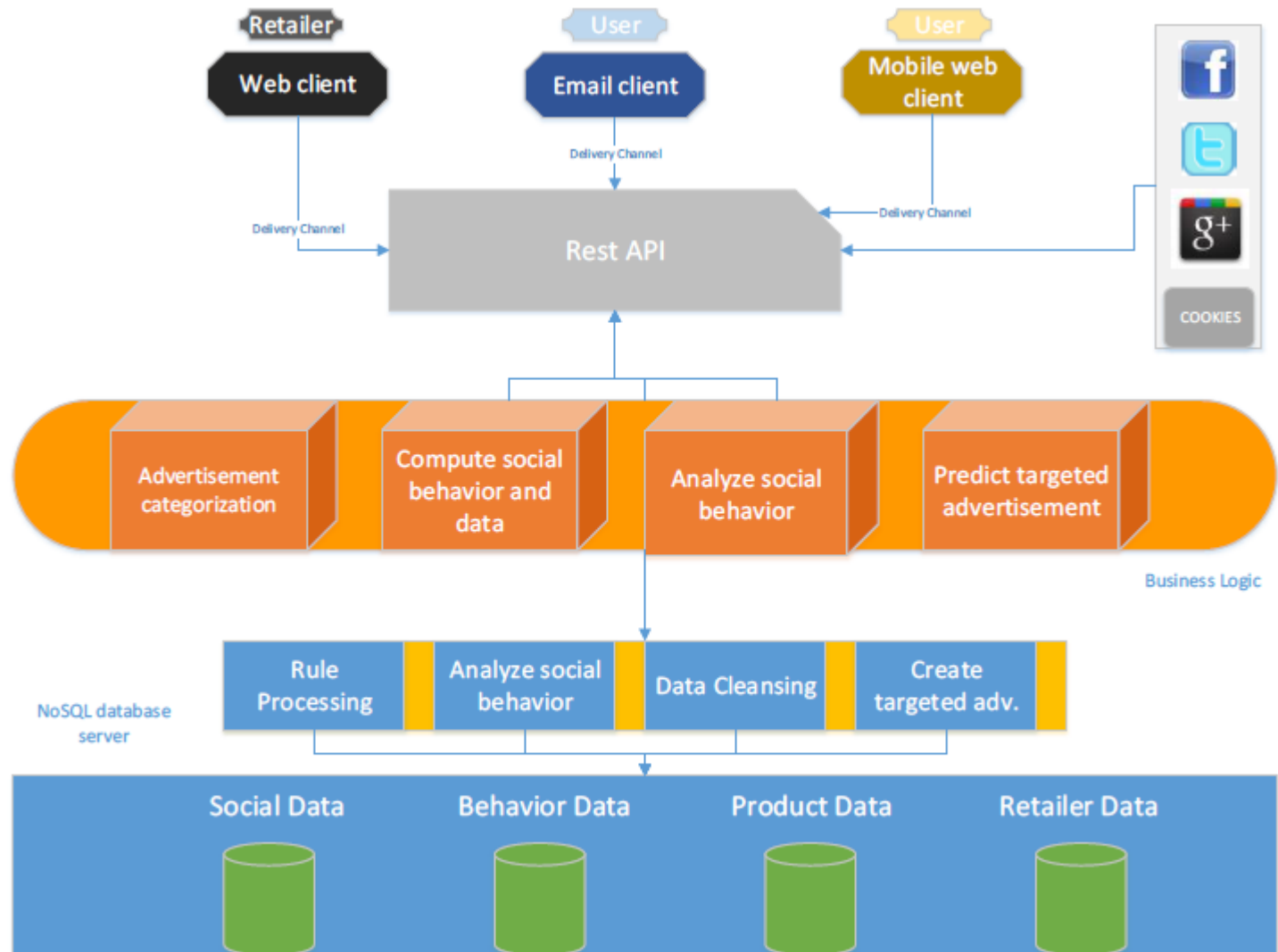
Data Mining – Kind of Data

Multi-dimensional data model



Source: Advanced analytics shape smart business outcomes for service providers - <http://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=TIS14129USEN&appname=wwwsearch>

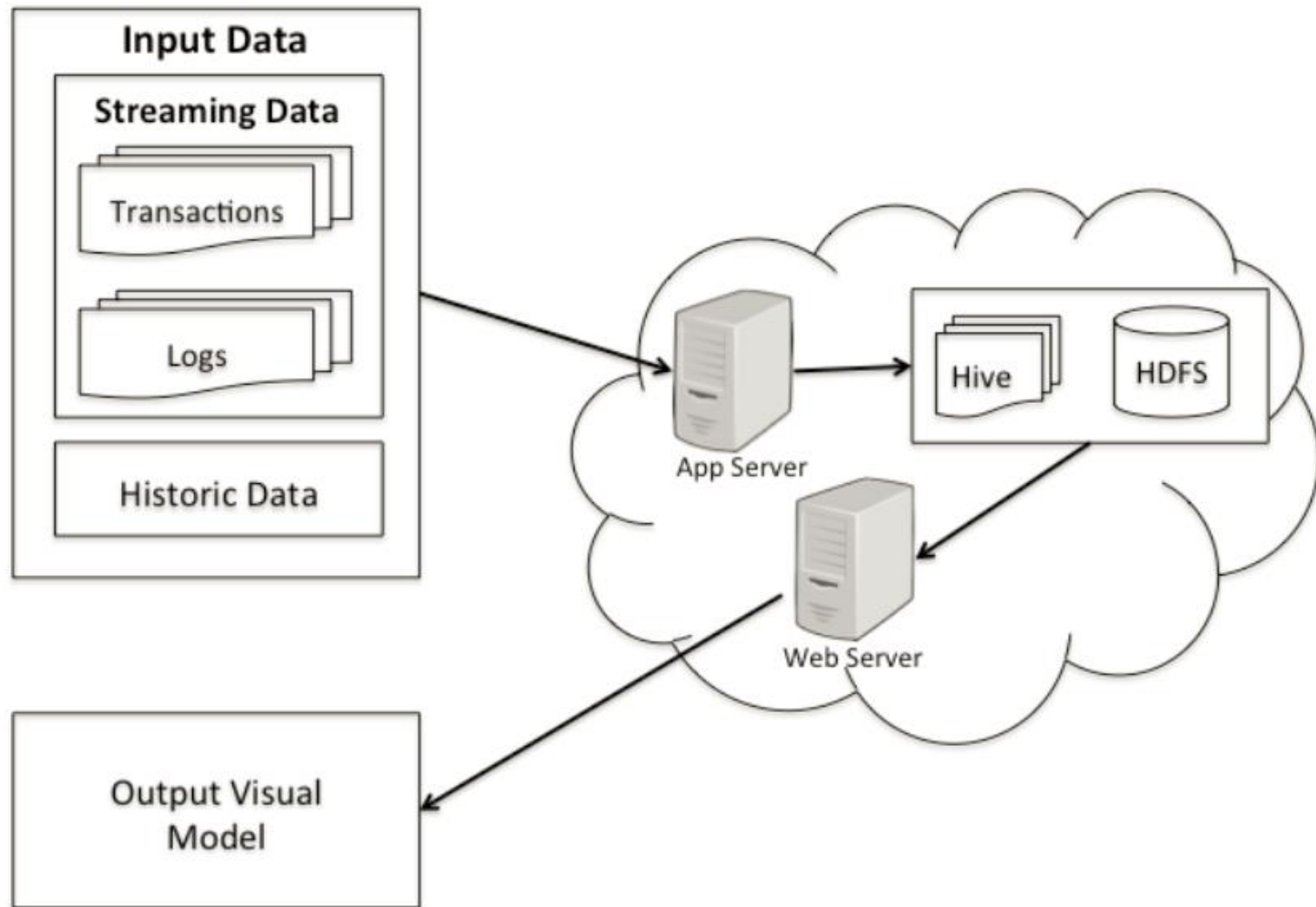
Data Mining – Kind of Data



Source: Advanced analytics shape smart business outcomes for service providers - <http://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=TIS14129USEN&appname=wwwse arch>



Data Mining – Kind of Data



Source: Advanced analytics shape smart business outcomes for service providers - <http://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=TIS14129USEN&appname=wwwsearch>

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

Data Mining – Kind of Data

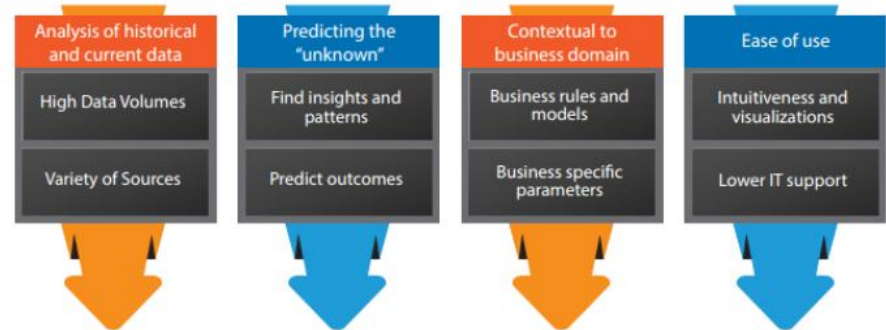
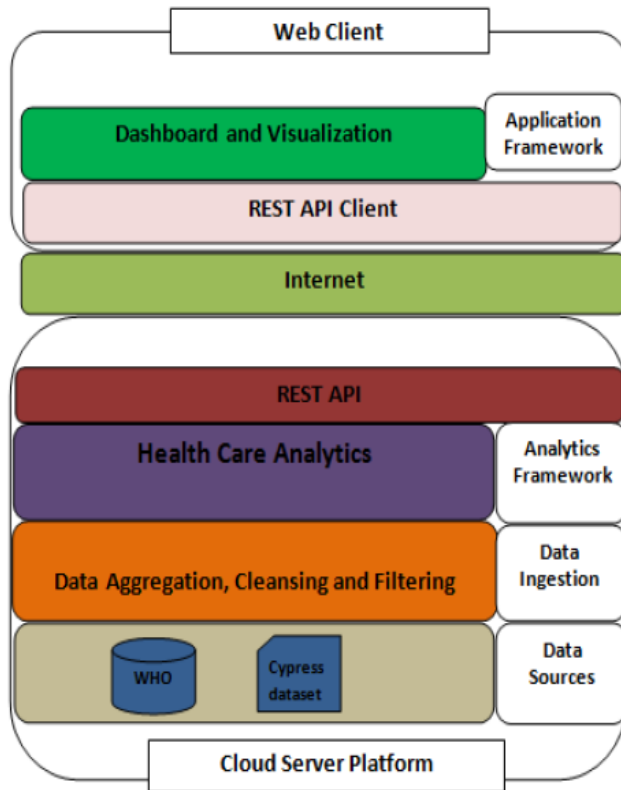
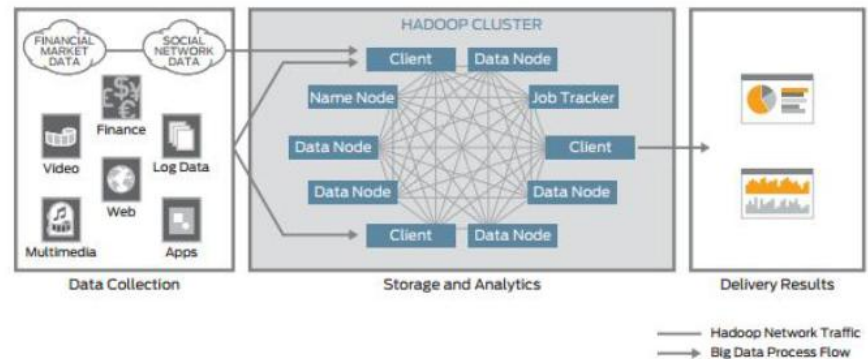


Figure 3. Characteristics of Predictive Analytics [10].

Big Data Analytics:



Source: Advanced analytics shape smart business outcomes for service providers - <http://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=TIS14129USEN&appName=wwwsearch>



TECHNIQUES FOR ANALYZING BIG DATA

Data Mining Functionalities

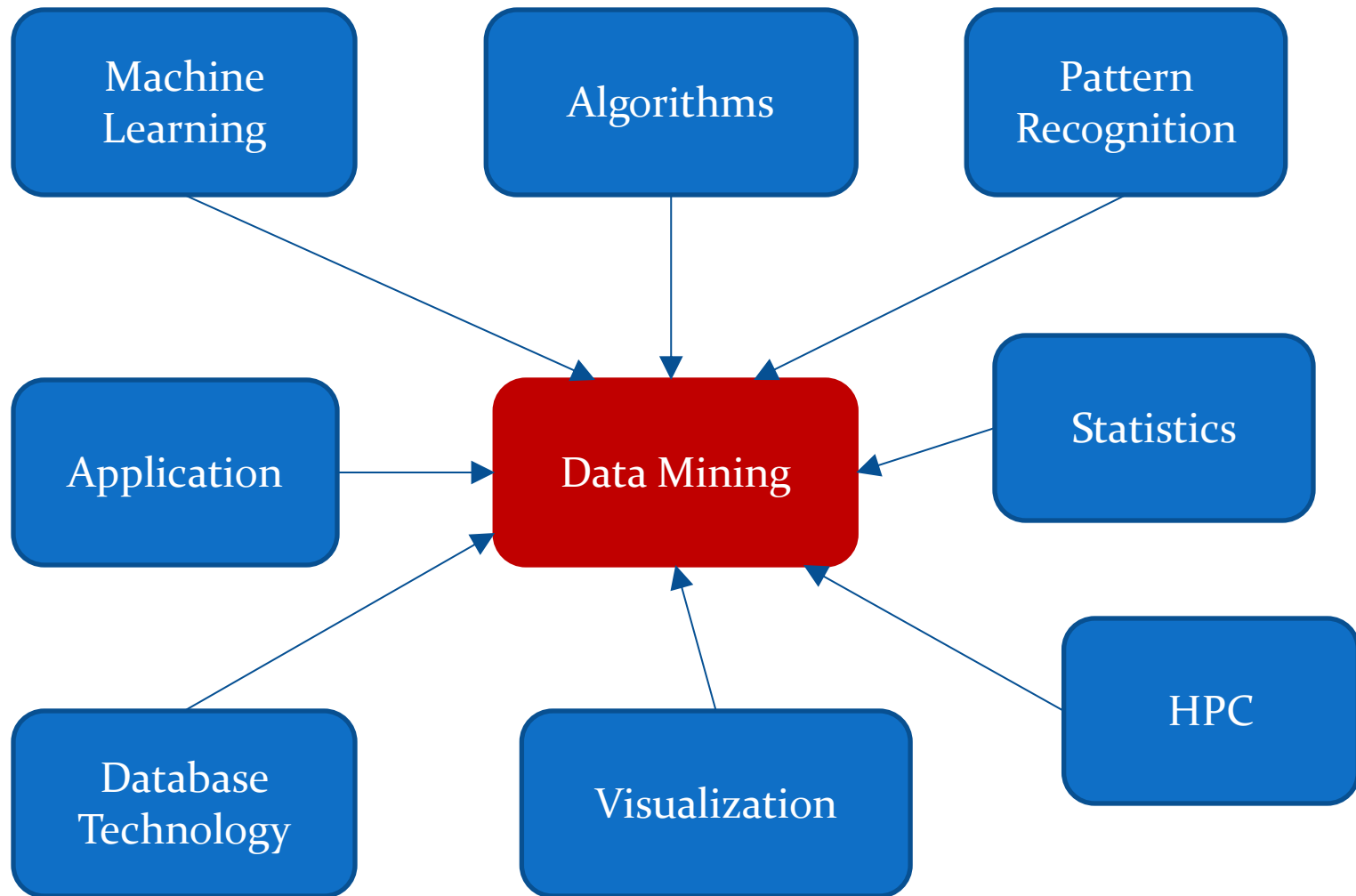
Data mining functionalities are used to specify the kind of patterns to be found in data mining tasks. In general, data mining tasks can be classified into two categories descriptive and predictive.

- **Descriptive mining** tasks characterize the general properties of the data in the database
- **Predictive mining** tasks perform inference on the current data in order to make predictions.

Source: An Introduction to Data Mining and Predictive Analytics



Data Mining: Confluence of Multiple Disciplines



Source: An Introduction to Data Mining and Predictive Analytics

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

TECHNIQUES FOR ANALYZING BIG DATA

Some of the techniques to analyze Big Data:

- Cluster Analysis
- Data Mining
- Predictive Modeling
- Regression
- Sentiment Analysis
- Signal Processing
- Time Series Analysis

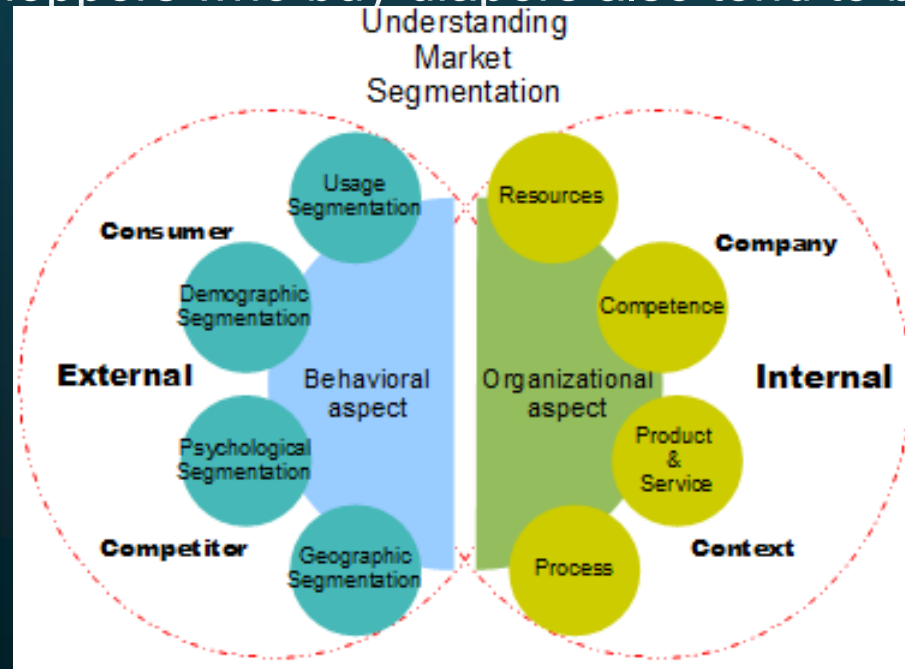
TECHNIQUES FOR ANALYZING BIG DATA

Data Mining. A set of techniques to extract patterns from large datasets by combining methods from statistics and machine learning with database management. These techniques include association rule learning, cluster analysis, classification, and regression. Applications include mining customer data to determine segments most likely to respond to an offer, mining human resources data to identify characteristics of most successful employees, or market basket analysis to model the purchase behavior of customers.



Data Mining Techniques

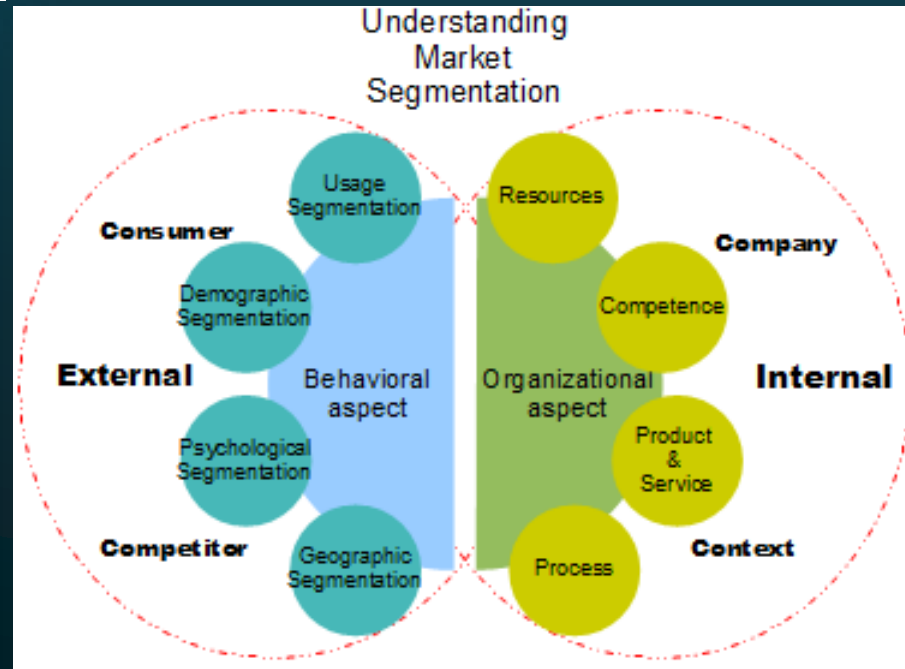
Association rule learning. A set of techniques for discovering interesting relationships, i.e., “association rules,” among variables in large databases. These techniques consist of a variety of algorithms to generate and test possible rules. One application is market basket analysis, in which a retailer can determine which products are frequently bought together and use this information for marketing (a commonly cited example is the discovery that many supermarket shoppers who buy diapers also tend to buy beer).



Source: 1. http://www.mckinsey.com/insights/business_technology/big_data_the_next_frontier_for_innovation

Data Mining Techniques

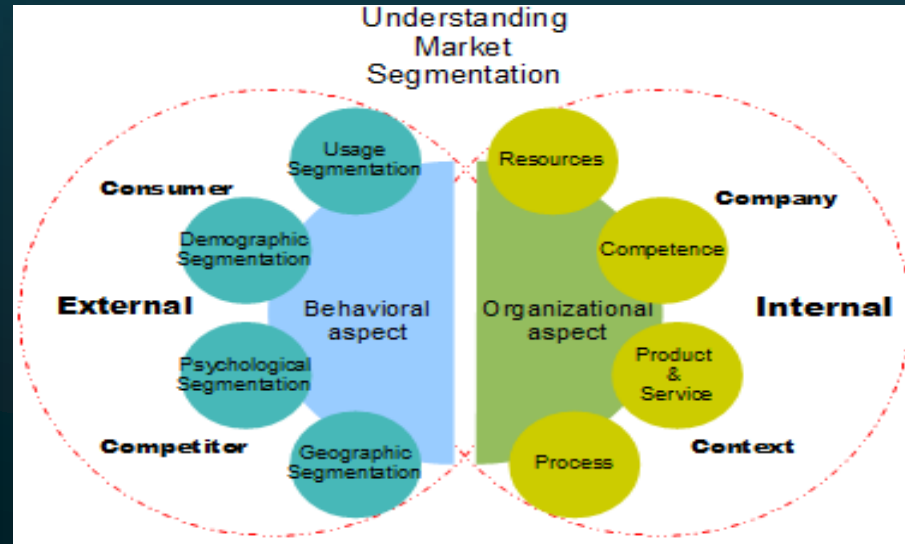
Cluster analysis. A statistical method for classifying objects that splits a diverse group into smaller groups of similar objects, whose characteristics of similarity are not known in advance. An example of cluster analysis is **segmenting consumers into self-similar groups for targeted marketing.** This is a type of unsupervised learning because training data are not used. This technique is in contrast to classification, a type of supervised learning. Used for data mining.



Source: 1. http://www.mckinsey.com/insights/business_technology/big_data_the_next_frontier_for_innovation

Data Mining Techniques

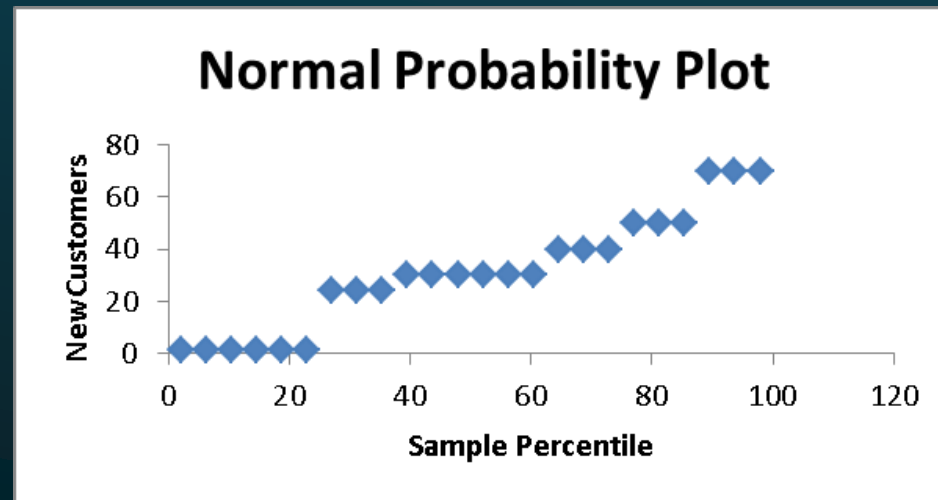
Classification. A set of techniques to identify the categories in which new data points belong, based on a training set containing data points that have already been categorized. One application is the prediction of segment-specific customer behavior (e.g., buying decisions, churn rate, consumption rate) where there is a clear hypothesis or objective outcome. These techniques are often described as supervised learning because of the existence of a training set; they stand in contrast to cluster analysis, a type of unsupervised learning.



Source: 1. http://www.mckinsey.com/insights/business_technology/big_data_the_next_frontier_for_innovation

Data Mining Techniques

Regression. A set of statistical techniques to determine how the value of the dependent variable changes when one or more independent variables is modified. Often used for **forecasting or prediction**. Examples of applications include forecasting sales volumes based on various market and economic variables or determining what measurable manufacturing parameters most influence customer satisfaction. Used for data mining.



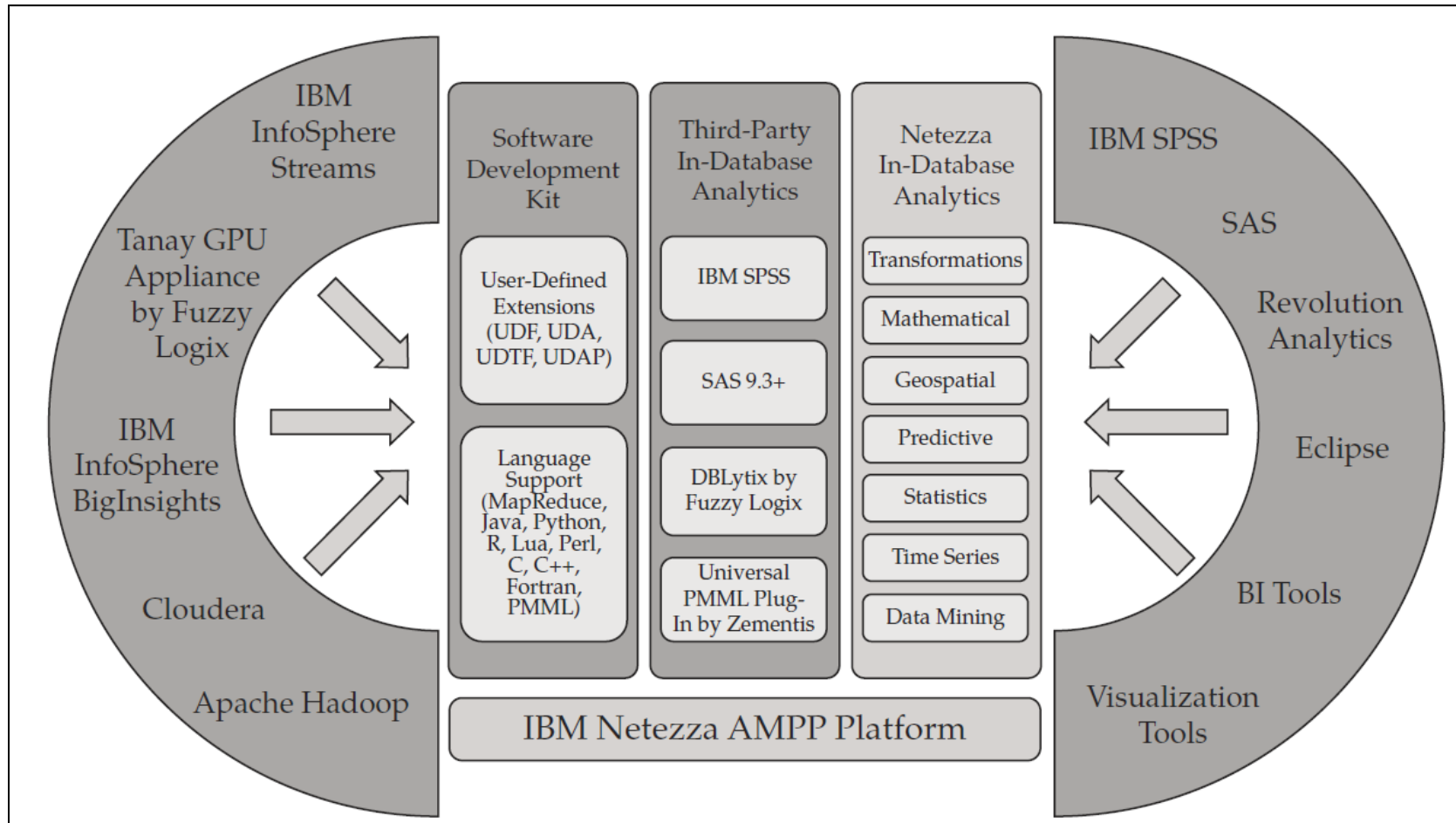
Why Data Mining?

- Data mining (knowledge discovery in databases):
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) information or patterns from data in large databases
- Alternative names and their “inside stories”:
 - Knowledge discovery(mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- What is not data mining?
 - (Deductive) query processing.
 - Expert systems or small ML/statistical programs

Source: Data Mining Concepts and Techniques Jaiwei Han and Machelaine Kambler



Data Mining System – IBM Netezza Platform



Source: Harness the power of the Big Data -

https://www14.software.ibm.com/webapp/iwm/web/signup.do?source=sw-infomgt&S_PKG=ov8257&S_TACT=109HF53W&S_CMP=big_data_site

Data Mining System – IBM Netezza Platform

Transformations	Execute in-database data transformations to realize significant performance gains.
Mathematical	Perform deep in-database mathematical calculations to leverage MPP processing.
Statistics	Calculate rich statistics without moving the data.
Time series	Create forecasts and identify trends using rich histories to improve model accuracy.
Data mining	Use more data, or all of the data, to discover new and emerging insights.
Predictive	Move from batch processing to near-real-time speed-of-thought analytics to predict with great accuracy and speed.
Geospatial	Implement location-based analytics on Big Data with immediate feedback.

Source: Harness the power of the Big Data -

https://www14.software.ibm.com/webapp/iwm/web/signup.do?source=sw-infomgt&S_PKG=ov8257&S_TACT=109HF53W&S_CMP=big_data_site



IBM SPSS Modeler

<http://www-01.ibm.com/software/analytics/spss/products/modeler/>



Source: <http://www-01.ibm.com/software/analytics/spss/products/modeler/>

IBM SPSS Modeler

The Data Mining Toolkit has operators to score several different types of data mining models in real time. Although data mining software such as SPSS requires multiple passes over data to build models, scoring can often be done on a record-by-record basis. The Data Mining Toolkit can score data mining models that are defined by the Predictive Model Markup Language (PMML) standard. This toolkit includes the following operators:

1. **Classification**: Supports Decision Trees, Naïve Bayes, and Logistic Regression algorithms that are used to classify tuples
2. **Clustering**: Supports Demographic Clustering and Kohonen Clustering algorithms that are used to assign tuples to a related group
3. **Regression**: Supports Linear Regression, Polynomial Regression, and Transform Regression algorithms that are used for predictive analytics
4. **Associations**: Supports association rules for predicting cause-and effect relationships

Source: <http://www-01.ibm.com/software/analytics/spss/products/modeler/>



Machine Learning

Arguably the single most significant factor driving the move toward truly advanced analytics will, in ABI Research's view, be machine learning and the way the vendors developing analytics software will utilize it. Namely, machine learning will allow analytical tools to autonomously or semi-autonomously spot patterns in the data, presenting suggestions and recommendations on what to do next, or making predictions on what may happen next. Understandably, this comes with its own caveats, but such fast tracks in data handling processes will, in turn, soften the cognitive and time constraints associated with the working methods that rely more on human labor and skills.

While elements of machine learning are being introduced by more or less all leading analytics vendors, the most dramatic examples of innovation can be seen among the start-ups that have built their own advanced analytics from the ground-up to leverage predictive algorithms. Meant to disrupt the query-centric status quo for interacting with data, these firms currently include, most notably, Skytree and Ayasdi. Of the two, Ayasdi is also putting a strong emphasis on the new ways of visualizing the findings derived from the machine learning engine by using topological data analysis. In this method, the relationships within the data sets are presented in the forms of shapes, colors, and network nodes, which will directly highlight the findings for the end user.

Source: UNLOCKING THE VALUE OF BIG DATA IN ENTERPRISES by Aapo Markkanen & Dan Shey



Web Mining

Discovering useful information from the content, structure and usage of the Web.

Web mining research overlaps substantially with other areas, including data mining, text mining, information retrieval, and Web retrieval. A possible classification of research in these areas is shown in the following table. The classification is based on two aspects: the purpose and the data sources. Retrieval research focuses on retrieving relevant, existing data or documents from a large database or document repository, while *mining* research focuses on discovering new information or knowledge in the data.

		Data/information sources		
		Any data	Textual data	Web-related data
Purpose	Retrieving known data or documents efficiently and effectively	Data Retrieval	Information Retrieval	Web Retrieval
	Finding new patterns or knowledge previously unknown	Data Mining	Text Mining	Web Mining

Source: <http://ai.arizona.edu/intranet/papers/WebMining.pdf>



Web Mining

Web mining is not simply the application of information retrieval and text mining techniques to Web pages; it also involves non-textual data such as Web server logs and other transaction-based data. From this point of view, Web retrieval and Web mining are considered overlapping areas, in which the main criterion for classification is the specific purpose of the application.

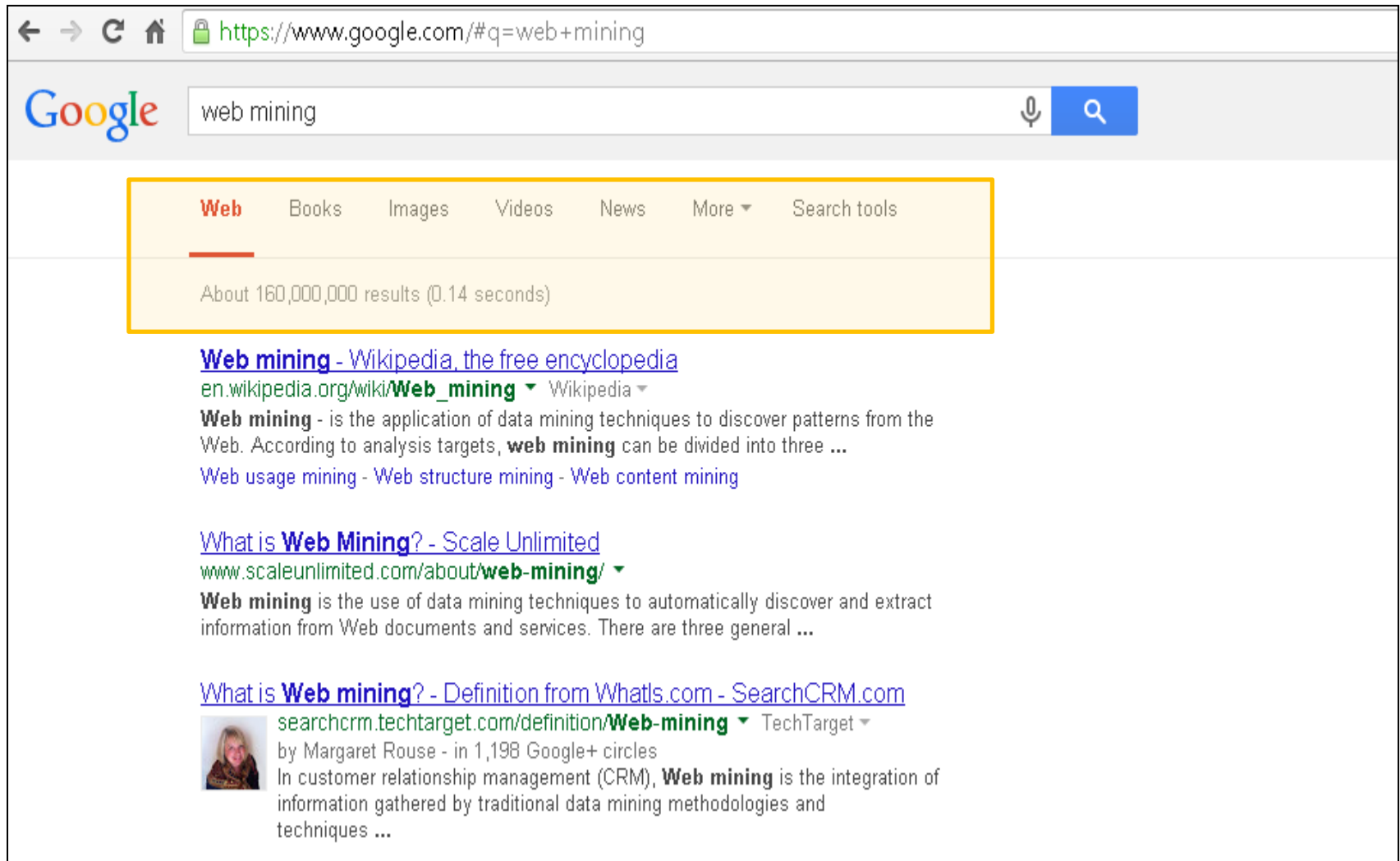
Web mining research can be divided into three categories:

- Web content mining
 - Text Mining for Web Documents
 - Intelligent Web Spiders
 - Multilingual Web Mining
 - Semantic Web
- Web structure mining, and
 - Citations
- Web usage mining
 - Pattern discovery & analysis
 - Personalization & Collarbarion

Source: <http://ai.arizona.edu/intranet/papers/WebMining.pdf>



Web Mining Applications

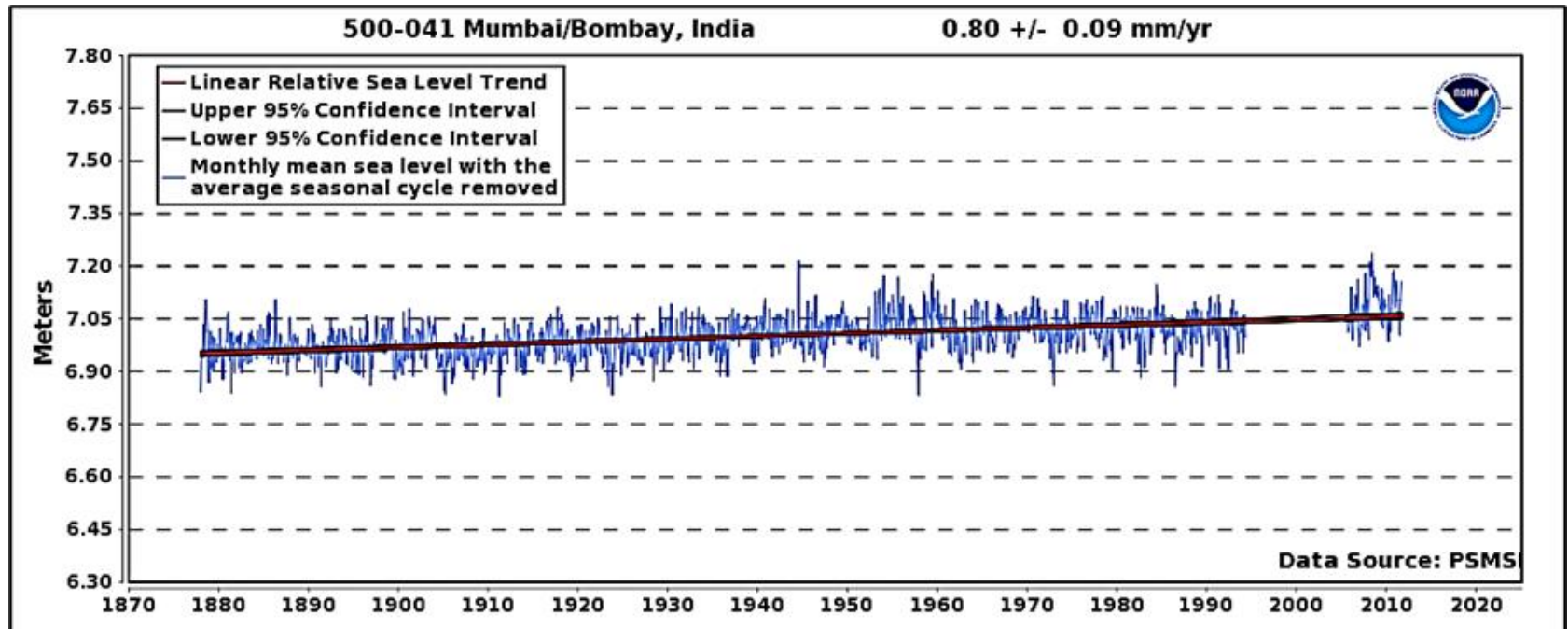


A screenshot of a Google search results page for the query "web mining". The browser's address bar shows the URL "https://www.google.com/#q=web+mining". The Google logo is on the left, and the search bar contains the text "web mining" with a microphone icon and a search button. Below the search bar, a yellow box highlights the search filters: "Web" (selected), "Books", "Images", "Videos", "News", "More", and "Search tools". Below this box, it says "About 160,000,000 results (0.14 seconds)". The search results list three entries:

- Web mining - Wikipedia, the free encyclopedia**
en.wikipedia.org/wiki/Web_mining - Wikipedia
Web mining - is the application of data mining techniques to discover patterns from the Web. According to analysis targets, **web mining** can be divided into three ...
[Web usage mining](#) - [Web structure mining](#) - [Web content mining](#)
- What is Web Mining? - Scale Unlimited**
www.scaleunlimited.com/about/web-mining/
Web mining is the use of data mining techniques to automatically discover and extract information from Web documents and services. There are three general ...
- What is Web mining? - Definition from WhatIs.com - SearchCRM.com**
 searchcrm.techtarget.com/definition/Web-mining - TechTarget
by Margaret Rouse - in 1,198 Google+ circles
In customer relationship management (CRM), **Web mining** is the integration of information gathered by traditional data mining methodologies and techniques ...

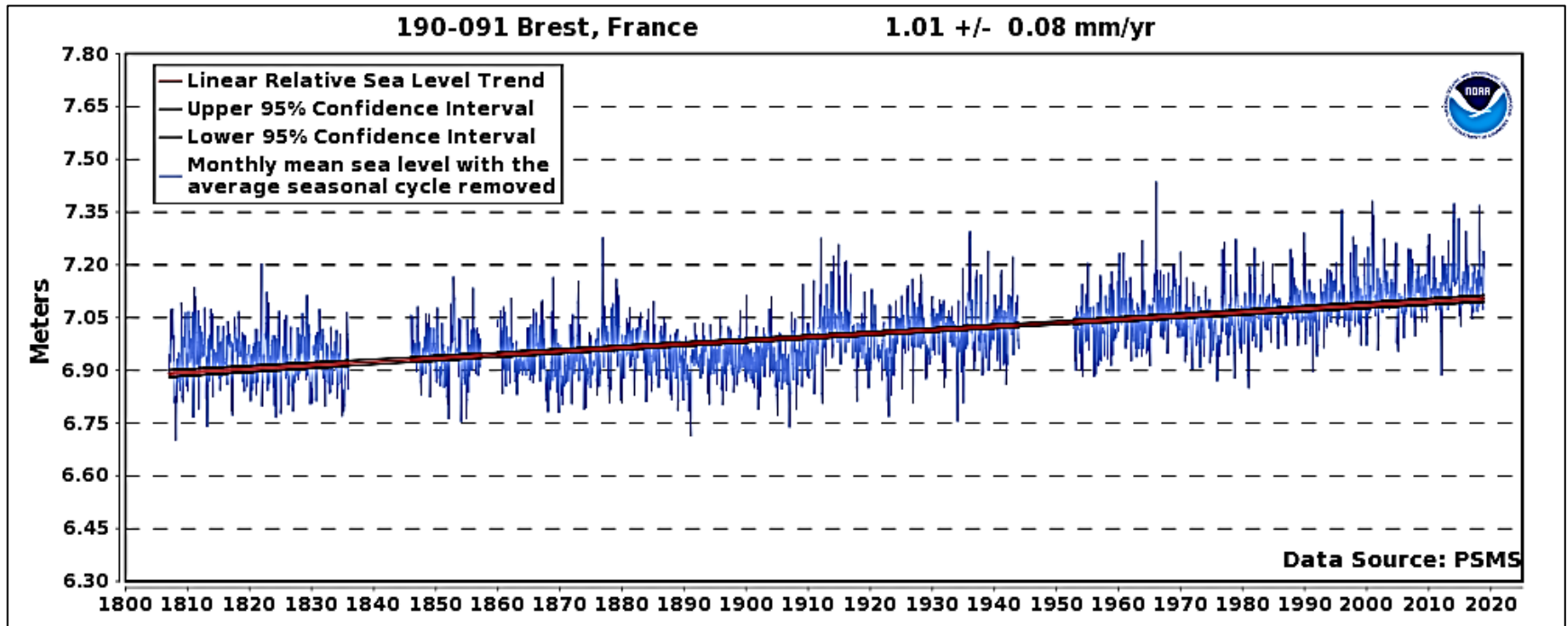


NOAA Datasets



Tides data - https://tidesandcurrents.noaa.gov/sltrends/sltrends_station.shtml?id=500-041

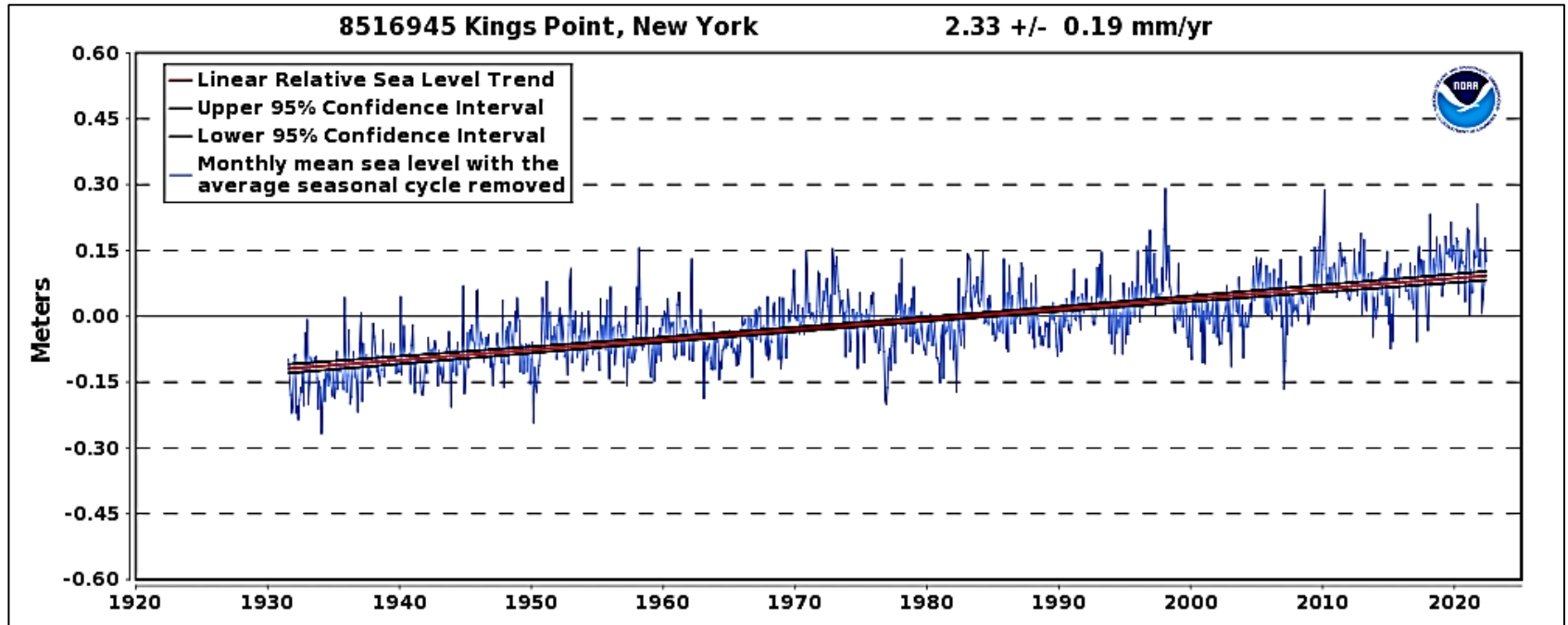
NOAA Datasets



Relative Sea Level Trend 190-091 Brest, France -

https://tidesandcurrents.noaa.gov/sltrends/sltrends_station.shtml?id=190-091

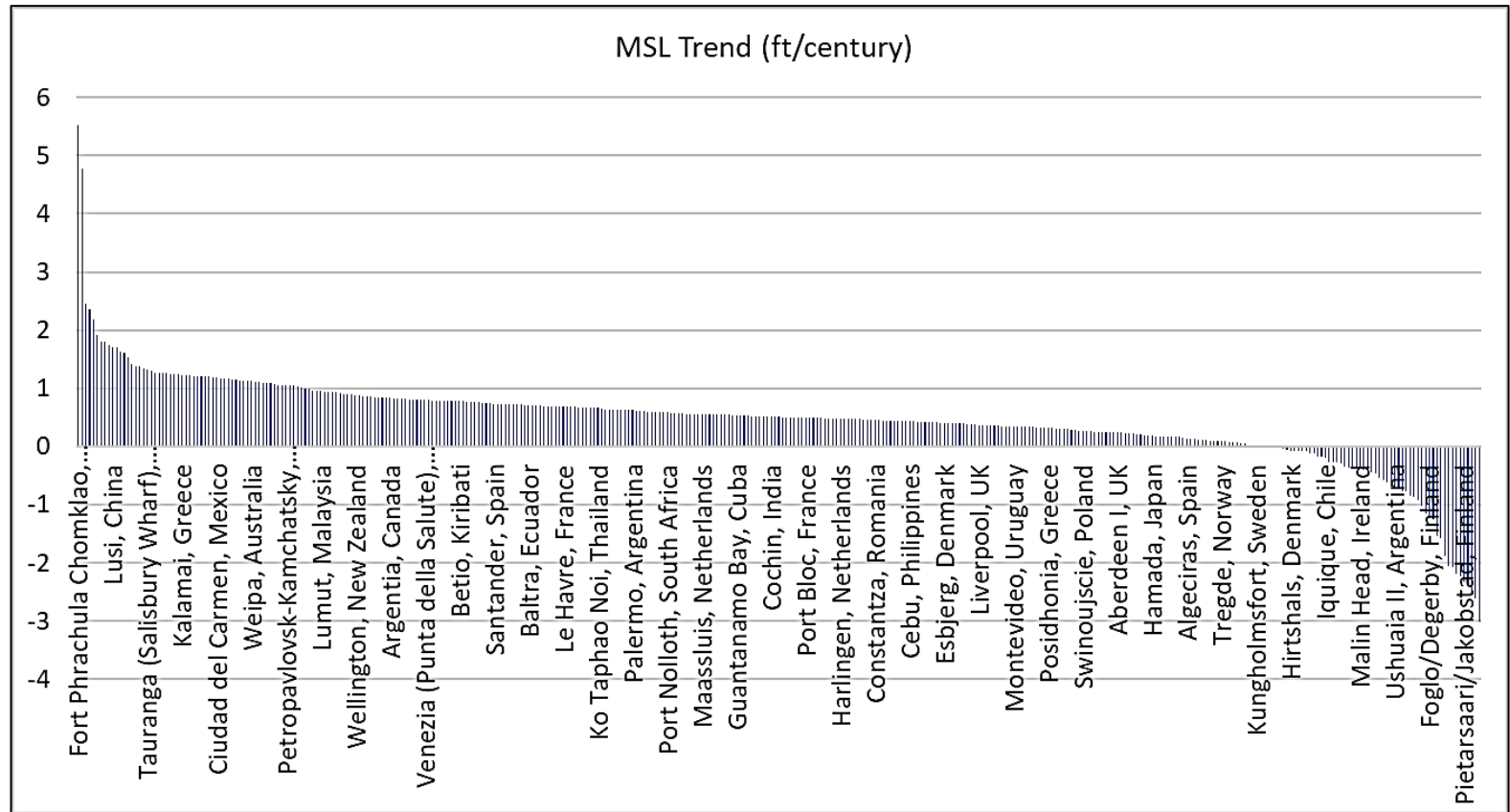
NOAA Datasets



Relative Sea Level Trend -

https://tidesandcurrents.noaa.gov/sltrends/sltrends_station.shtml?id=8516945

NOAA Datasets



Global Linear Relative Sea Level (RSL) trends and 95% Confidence Intervals (CI) in mm/year and in ft/century -

<https://tidesandcurrents.noaa.gov/sltrends/mslGlobalTrendsTable.html>

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- Three proven methods to achieve a higher ROI from data minings, <http://public.dhe.ibm.com/common/ssi/ecm/en/imw14282usen/IMW14282USEN.PDF>
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